Dirichlet Series and L-functions

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1 Dirichlet Series

Throughout, $s = \sigma + it$ and $s_0 = \sigma_0 + it_0$ will stand for complex variables with σ, σ_0, t, and t_0 real. We will also use the Mellin transform and its inversion theorem.

1.1 Convergence Properties

An (analytic) **Dirichlet series** $D(s)$ is a function of the form

$$D(s) = \sum_{n \geq 1} \frac{a(n)}{n^s},$$

for some complex coefficients $a(n)$. Our first aim is to understand where Dirichlet series converge and where they converge absolutely. It does not take much for $D(s)$ to converge uniformly in a sector.

Theorem 1.1. *Suppose $D(s)$ is a Dirichlet series that converges at $s_0 = \sigma_0 + it_0$. Then for any $H > 0$, $D(s)$ converges uniformly in the sector*

$$\{s \in \mathbb{C} : \sigma \geq \sigma_0 \text{ and } |t - t_0| \leq H(\sigma - \sigma_0)\}.$$

In particular, $D(s)$ converges in the half-plane $\sigma > \sigma_0$.

Proof. Write $R(u) = \sum_{n \geq u} \frac{a(n)}{n^{s_0}}$ for the tail of $D(s_0)$ so that

$$\frac{a(n)}{n^{s_0}} = (R(n) - R(n+1)).$$

Then for positive integers N and M with $M \leq N$, summation by parts implies

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = R(M)M^{s_0 - s} - R(N+1)(N+1)^{s_0 - s} - \sum_{M \leq n \leq N} R(n+1)(n^{s_0 - s} - (n+1)^{s_0 - s}).$$

We will express the sum on the right-hand side as an integral. To do this, observe that

$$n^{s_0 - s} - (n+1)^{s_0 - s} = -(s_0 - s) \int_n^{n+1} u^{s_0 - s - 1}\, du.$$

As $R(u)$ is constant on the interval $(n, n+1]$ a short computation shows

$$\sum_{M \leq n \leq N} R(n+1)(n^{s_0 - s} - (n+1)^{s_0 - s}) = -(s_0 - s) \int_M^{N+1} R(u) u^{s_0 - s - 1}\, du,$$

Whence

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = R(M)M^{s_0 - s} - R(N+1)(N+1)^{s_0 - s} + (s_0 - s) \int_M^{N+1} R(u) u^{s_0 - s - 1}\, du.$$

As $D(s)$ converges at $s = s_0$, we can choose M sufficiently large such that $|R(u)| < \varepsilon$ for $u \geq M$. It follows that $|R(u)u^{s_0 - s}| < \varepsilon$ for such u provided s in the desired sector. For such s, we have

$$|s - s_0| \leq (\sigma - \sigma_0) + |t - t_0| \leq (H+1)(\sigma - \sigma_0).$$

These estimates together imply

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = O\left(2\varepsilon + \varepsilon(H+1)(\sigma - \sigma_0) \int_M^{N+1} u^{\sigma_0 - \sigma - 1}\, du\right).$$

Since the integral is $O\left(\frac{1}{\sigma - \sigma_0}\right)$, the sum is $o(1)$ for s in the desired sector. The first statement follows by uniform Cauchy's criterion. Taking the limit as $H \to \infty$ proves the second statement. $\square$

We will want to keep track of where Dirichlet series converge absolutely. Let σ_c be the infimum of all σ for which $D(s)$ converges. We call σ_c the **abscissa of convergence** of $D(s)$. Similarly, let σ_a be the infimum of all σ for which $D(s)$ converges absolutely. We call σ_a the **abscissa of absolute convergence** of $D(s)$. As the summands of $D(s)$ are holomorphic, convergence is locally absolutely uniform for $\sigma > \sigma_a$ (actually uniform in sectors) and so $D(s)$ is holomorphic in this half-plane.

The abscissas σ_c and σ_a act as the boundaries of convergence and absolute convergence respectively. Anything can happen on the lines $\sigma = \sigma_c$ and $\sigma = \sigma_a$, but to the right of them we have convergence and absolute convergence of $D(s)$ respectively. It turns out that σ_a is never far from σ_c provided σ_c is finite.

Theorem 1.2. *If $D(s)$ is a Dirichlet series with finite abscissa of convergence σ_c then*

$$\sigma_c \leq \sigma_a \leq \sigma_c + 1.$$

Proof. The first inequality is clear since absolute convergence implies convergence. For the second inequality, the terms $a(n)n^{-(\sigma_c+\varepsilon)}$ tend to zero as $n \to \infty$ because $D(s)$ converges at $s = \sigma_c + \varepsilon$. Therefore $a(n) \ll_\varepsilon n^{\sigma_c+\varepsilon}$ and so $D(s)$ is absolutely convergent at $s = \sigma_c + 1 + 2\varepsilon$. This means $\sigma_a \leq \sigma_c + 1 + 2\varepsilon$. Taking the limit as $\varepsilon \to 0$ gives the second inequality. □

We now turn to the question of uniqueness of Dirichlet series. In particular, we would like to show that Dirichlet series are uniquely determined by their coefficient as this would allow us compare coefficient analogous to that of Taylor series. This is indeed possible.

Proposition 1.3. *Suppose $D(s)$ is a Dirichlet series with finite abscissa of convergence σ_c such that*

$$D(s) = \sum_{n\geq 1} \frac{a(n)}{n^s} \quad \text{and} \quad D(s) = \sum_{n\geq 1} \frac{b(n)}{n^s}.$$

Then $a(n) = b(n)$.

Proof. Set $c(n) = a(n) - b(n)$ so that it suffices to prove $c(n) = 0$. Take $\sigma > \sigma_a$ so that $D(s)$ converges uniformly. Letting $\sigma \to \infty$, we have

$$\lim_{\sigma\to\infty} D(\sigma) = a(1) \quad \text{and} \quad \lim_{\sigma\to\infty} D(\sigma) = b(1),$$

upon interchanging the limit and the sum by uniform convergence. Whence $c(1) = 0$. Assume by induction that $c(n) = 0$ for $n \leq N$ and consider the Dirichlet series

$$\sum_{n\geq 1} \frac{c(n)}{n^s}.$$

Its abscissa of absolute convergence is σ_a. As

$$\sum_{n\leq N} \frac{c(n)}{n^s} = 0,$$

by assumption, it follows that

$$c(N+1) = -\sum_{n>N} c(n)\left(\frac{N}{n}\right)^{\sigma}.$$

Take $\sigma > \sigma_a$ so that sum on the right-hand side converges uniformly. Letting $\sigma \to \infty$, we have

$$\lim_{\sigma\to\infty}\left(-\sum_{n>N} c(n)\left(\frac{N}{n}\right)^{\sigma}\right) = 0,$$

upon interchanging the limit and sum by uniform convergence. Hence $c(N+1) = 0$ which completes the proof. □

We will now introduce several resulting concerning the growth and average growth of the coefficients of a Dirichlet series. For legibility, it will be useful to introduce some notation. If $D(s)$ is a Dirichlet series with coefficients $a(n)$ then for $X > 0$, we set

$$A(X) = \sum_{n\leq X} a(n) \quad \text{and} \quad |A|(X) = \sum_{n\leq X} |a(n)|.$$

These are the partial sums of the coefficients $a(n)$ and their absolute values up to X respectively. Many of our results will be in terms of these sums. Our first result shows that the coefficients of a Dirichlet series grown at most polynomially provided the Dirichlet series converges absolutely at some point.

Proposition 1.4. *Suppose $D(s)$ is a Dirichlet series with coefficients $a(n)$ that converges absolutely at $s = \alpha$ for some real α. Then*

$$a(n) \ll n^{\alpha}.$$

In particular, if $D(s)$ has finite abscissa of absolute convergence σ_a then

$$a(n) \ll_{\varepsilon} n^{\sigma_a+\varepsilon}.$$

Proof. Write

$$|a(n)| \leq n^{\alpha}\sum_{n\geq 1}\frac{|a(n)|}{n^{\alpha}}.$$

The sum is $O(1)$ because $D(s)$ is absolutely convergent at $\sigma = \alpha$. The first estimate follows at once. The second is immediate from the first and the definition of the abscissa of absolute convergence upon setting $\alpha = \sigma_a + \varepsilon$. □

It is natural to be curious about the converse, namely, is it possible to determine an upper bound on the abscissa of absolute convergence if we know a polynomial bound for the coefficients. This is indeed possible.

Proposition 1.5. *Suppose $D(s)$ is a Dirichlet series whose coefficients satisfy the estimate $a(n) \ll n^{\alpha}$ for some real α. Then the abscissa of absolute convergence satisfies $\sigma_a \leq \alpha + 1$.*

Proof. Let $\sigma > \alpha + 1$. As

$$\sum_{n\geq 1} \frac{|a(n)|}{n^\sigma} \ll \sum_{n\geq 1} \frac{1}{n^{\sigma-\alpha}},$$

and the latter series converges, the proof is complete. $\square$

Let us now turn to the same questions but using averages of the coefficients. Our first result is analogous to Proposition 1.4.

Proposition 1.6. *Suppose $D(s)$ is a Dirichlet series with coefficients $a(n)$ that converges absolutely at $s = \alpha$ for some real α. Then*

$$A(X) \ll X^\alpha \quad \text{and} \quad |A|(X) \ll X^\alpha.$$

In particular,

$$A(X) \ll_\varepsilon X^{\sigma_a+\varepsilon} \quad \text{and} \quad |A|(X) \ll_\varepsilon X^{\sigma_a+\varepsilon}.$$

Proof. Write

$$\sum_{n\leq X} |a(n)| \leq X^\alpha \sum_{n\geq 1} \frac{|a(n)|}{n^\alpha}.$$

The sum is $O(1)$ because $D(s)$ is absolutely convergent at $s = \alpha$. As $A(x) \ll |A|(x)$, the first statement follows at once. The second is immediate from the first and the definition of the abscissa of absolute convergence upon setting $\alpha = \sigma_a + \varepsilon$. $\square$

It turns out that if $A(X)$ is bounded then σ_c is negative.

Proposition 1.7. *Suppose $D(s)$ is a Dirichlet series and that $A(X) \ll 1$. Then $\sigma_c \leq 0$.*

Proof. Let $\sigma > 0$. Abel summation gives

$$\sum_{n\leq X} \frac{a(n)}{n^s} = \frac{A(X)}{X^s} + \int_1^X A(u)u^{-(s-1)}\, du.$$

As $A(X) \ll 1$, take the limit as $X \to \infty$ to obtain

$$D(s) = s\int_1^\infty A(u)u^{-(s+1)}\, du.$$

This expresses $D(s)$ as an integral. Direct evaluation shows that the integral is finite. Therefore $D(s)$ converges for $\sigma > 0$ which means $\sigma_c \leq 0$. $\square$

Unfortunately, it is not often the case that $A(X)$ is bounded as this is quite a strong condition for most Dirichlet series. Fortunately, we can still obtain nice result analogous to that of Proposition 1.5 if we assume $|A|(X)$ grows at most polynomially.

Proposition 1.8. *Suppose $D(s)$ is a Dirichlet series such that $|A|(X) \ll X^\alpha$ for some $\alpha \geq 0$. Then the abscissa of absolute convergence satisfies $\sigma_a \leq \alpha$.*

Proof. Let $\sigma > \alpha$. It suffices to show convergence of the series

$$\sum_{n\geq 1}\frac{|a(n)|}{n^\sigma}.$$

Abel summation gives

$$\sum_{n\leq X}\frac{|a(n)|}{n^\sigma} = \frac{|A|(X)}{X^\sigma} + \sigma\int_1^X |A|(u)u^{-(\sigma+1)}\,du.$$

In view of the bound $|A|(X) \ll X^\alpha$, take the limit as $X \to \infty$ to obtain

$$\sum_{n\geq 1}\frac{|a(n)|}{n^\sigma} = \sigma\int_1^\infty |A|(u)u^{-(\sigma+1)}\,du.$$

Direct evaluation shows that the integral is bounded. Therefore our series is bounded and hence must converge. $\square$

Unfortunately, Proposition 1.8 does not immediately imply a sharper upper bound for the abscissa of absolute convergence than that of Proposition 1.5. This is because if $a(n) \ll n^\alpha$ then the trivial bounds for $A(X)$ and $|A(X)|$ are

$$A(X) \ll X^{\alpha+1} \quad \text{and} \quad |A|(X) \ll X^{\alpha+1}.$$

In particular, using the second bound in Proposition 1.8 will give $\sigma_a \leq \alpha+1$. So if we want to obtain sharper upper bounds for the abscissa of absolute convergence then we must improve the polynomial bound for the coefficients directly. Unfortunately, this is often a daunting task in practice especially if the Dirichlet series is connected to a deep algebraic or arithmetic object. However, if we assume that the coefficients are nonnegative then **Landau's theorem** provides a way of locating the abscissa of absolute convergence exactly:

Theorem (Landau's theorem). *Suppose $D(s)$ is a Dirichlet series with nonnegative coefficients $a(n)$ and finite abscissa of absolute convergence σ_a. Then σ_a is a singularity of $D(s)$.*

Proof. Replacing $a(n)$ by $a(n)n^{-\sigma_a}$, if necessary, we may assume $\sigma_a = 0$. Now suppose to the contrary that $D(s)$ was holomorphic at $s = 0$. Then $D(s)$ is holomorphic in the domain

$$\mathcal{D} = \{s \in \mathbb{C} : \sigma_a > 0\} \cup \{s \in \mathbb{C} : |s| < \delta\},$$

for some $\delta > 0$. Let $P(s)$ be the power series expansion of $D(s)$ at $s = 1$. Then

$$P(s) = \sum_{k\geq 0}\frac{c_k}{k!}(s-1)^k,$$

where

$$c_k = \sum_{n\geq 1}\frac{a(n)(-\log(n))^k}{n},$$

upon differentiating $D(s)$ termwise. The radius of convergence of $P(s)$ is the distance from $s=1$ to the nearest singularity of $P(s)$. Since $P(s)$ is holomorphic on $\mathcal{D}$, the closest possible singularities are at $s=\pm i\delta$. Therefore, the radius of convergence is at least $\sqrt{1+\delta^2}$. Write $\sqrt{1+\delta^2}=1+\delta'$ for some $\delta'>0$. Then for $|s-1|<1+\delta'$, $P(s)$ is holomorphic and can be expressed as

$$P(s)=\sum_{k\geq 0}\frac{(s-1)^k}{k!}\sum_{n\geq 1}\frac{a(n)(-\log(n))^k}{n}.$$

Now choose s in this region to be real with with $-\delta'<s<1$. Then this double sum is a sum of positive terms by assumption of the $a(n)$ being nonnegative. As $P(s)$ is necessarily absolutely convergent, interchanging the sums and a short computation shows

$$P(s)=D(s).$$

As $-\delta'<s<1$ and $D(s)$ has nonnegative coefficients, it converges absolutely for some $s<0$. This contradicts $\sigma_a=0$. $\square$

There is a very important distinction in Landau's theorem that abscissa of absolute convergence is a singularity and not just a point where the Dirichlet series does not converge. Indeed, this means that if $D(s)$ could be analytically continued to a region containing $\sigma=\sigma_a$ then the continuation must have a pole at $s=\sigma_a$. In particular, the abscissa of absolute convergence would then be a pole with the largest possible real part. This signals that the singularity is an inherent property of the analytic continuation and not one of the representation of the function as a Dirichlet series.

1.2 Euler Products

Generally speaking, if the coefficients $a(n)$ are chosen at random, $D(s)$ is not guaranteed to good properties outside of absolute convergence in some half-plane (provided it converges at a point). However, many Dirichlet series of interest will have coefficients that are multiplicative. These Dirichlet series admits product expressions.

Proposition 1.9. *Suppose $D(s)$ is a Dirichlet series with finite abscissa of absolute convergence σ_a and whose coefficients $a(n)$ are multiplicative. Then*

$$D(s)=\prod_p\left(\sum_{k\geq 0}\frac{a(p^k)}{p^{ks}}\right),$$

for $\sigma>\sigma_a$. If the prime power coefficients satisfy

$$a(p^k)=\sum_{j_1\leq\cdots\leq j_k}\alpha_{j_1}(p)\cdots\alpha_{j_k}(p),$$

for some $\alpha_1(p),\ldots,\alpha_d(p)\in\mathbb{C}$ and positive integer d, then the product takes the form

$$D(s)=\prod_p(1-\alpha_1(p)p^{-s})^{-1}\cdots(1-\alpha_d(p)p^{-s})^{-1}.$$

Proof. Let $\sigma > \sigma_a$ and consider the series

$$\sum_{k\geq 0} \frac{a(p^k)}{p^{ks}}.$$

This series converges absolutely because $D(s)$ does. Now let N be a positive integer. By absolute convergence and the fundamental theorem of arithmetic,

$$\prod_{p\leq N}\left(\sum_{k\geq 0} \frac{a(p^k)}{p^{ks}}\right) = \sum_{n\leq N} \frac{a(n)}{n^s} + \sum_{n>N}{}^{*} \frac{a(n)}{n^s},$$

where the $*$ indicates that we are summing over only those additional terms $\frac{a(n)}{n^s}$ that appear in the expanded product. As $N \to \infty$, the first sum on the right-hand side tends to $D(s)$ and the second sum tends to zero because it is part of the tail of $D(s)$. This proves that the product converges and is equal to $D(s)$. If the prime power coefficients are of the form

$$a(p^k) = \sum_{j_1\leq\cdots\leq j_k} \alpha_{j_1}(p)\cdots\alpha_{j_k}(p),$$

then

$$\sum_{k\geq 0} \frac{a(p^k)}{p^{ks}} = (1-\alpha_1(p)p^{-s})^{-1}\cdots(1-\alpha_d(p)p^{-s})^{-1},$$

from which the product formula follows. □

The representation

$$D(s) = \prod_p \left(\sum_{k\geq 0} \frac{a(p^k)}{p^{ks}}\right),$$

the called the **Euler product** of $D(s)$. By Proposition 1.9, multiplicativity of the coefficients is enough to ensure that the Euler product exists and is equal to the Dirichlet series in the half-plane of absolute convergence. If the Euler product takes the form

$$D(s) = \prod_p (1-\alpha_1(p)p^{-s})^{-1}\cdots(1-\alpha_d(p)p^{-s})^{-1},$$

then it is said to be of **degree** d. The special case of complete multiplicative coefficients corresponds to degree 1 Euler products as

$$D(s) = \prod_p (1-a(p)p^{-s})^{-1}.$$

In any case, the existence of $\alpha_1(p),\ldots,\alpha_d(p) \in \mathbb{C}$ with

$$a(p^k) = \sum_{j_1\leq\cdots\leq j_k} \alpha_{j_1}(p)\cdots\alpha_{j_k}(p),$$

guarantees that the Dirichlet series has an Euler product of degree d.

Remark 1.10. Replacing $D(s)$ with its absolute series in Proposition 1.9 shows that

$$\sum_{n\geq 1}\frac{|a(n)|}{n^{\sigma}} = \prod_{p}\left(\sum_{k\geq 0}\frac{|a(p^k)|}{p^{k\sigma}}\right)$$

for $\sigma > \sigma_a$. Under the stronger assumption of an Euler product of degree d this identity becomes

$$\sum_{n\geq 1}\frac{|a(n)|}{n^{\sigma}} = \prod_{p}(1-|\alpha_1(p)|p^{-\sigma})^{-1}\cdots(1-|\alpha_d(p)|p^{-\sigma})^{-1}.$$

Either of these equalities is stronger than mere absolute convergence of the infinite product since each factor is replaced with its absolute series not just the absolute value of the series. In particular, this implies that the Euler product converges locally absolutely uniformly in the same region that the Dirichlet series does.

If $D(s)$ admits a Euler product, we write $D^{(N)}(s)$ to denote the Dirichlet series with the factors $p \mid N$ in the Euler product removed. This means

$$D^{(N)}(s) = D(s)\prod_{p\mid N}\left(\sum_{k\geq 0}\frac{a(p^k)}{p^{ks}}\right).$$

Dually, we let $D_{(N)}(s)$ denote the Dirichlet series consisting only of the factors $p \mid N$ in the Euler product. This means

$$D_{(N)}(s) = \prod_{p\mid N}\left(\sum_{k\geq 0}\frac{a(p^k)}{p^{ks}}\right).$$

With this notation, we have the relationship

$$D(s) = D^{(N)}(s)D_{(N)}(s).$$

1.3 Dirichlet Convolution

We now turn to discussing how Dirichlet series behave with respect to products. Let $D_f(s)$ and $D_g(s)$ be the Dirichlet series defined by

$$D_f(s) = \sum_{n\geq 1}\frac{f(n)}{n^s} \quad\text{and}\quad D_g(s) = \sum_{n\geq 1}\frac{g(n)}{n^s},$$

for some arithmetic functions f and g. Let $f * g$ be the Dirichlet convolution of f and g. A short computation shows

$$D_f(s)D_g(s) = D_{f*g}(s).$$

In other words, $D_f(s)D_g(s)$ is again a Dirichlet series whose coefficients are given by the Dirichlet convolution of f and g. This relation also shows that $D_{f*g}(s)$ converges absolutely wherever both $D_f(s)$ and $D_g(s)$ do. Since Dirichlet convolution preserves multiplicativity,

$D_{f*g}(s)$ will have multiplicative coefficients if both $D_f(s)$ and $D_g(s)$ do. Moreover, from the Möbius inversion formula we immediately find that

$$D_g(s) = D_{f*\mathbf{1}}(s),$$

if and only if

$$D_f(s) = D_{g*\mu}(s).$$

These identities can be used to compute the Dirichlet series for many types of arithmetic functions.

1.4 Perron Formulas

With Mellin inversion, it is possible to relate the sums of coefficients of a Dirichlet series to an integral of its associated Dirichlet series. Such formulas are desirable because they allow for the examination of these sums by methods in complex analysis. First, we setup some general notation. Let $D(s)$ be a Dirichlet series with coefficients $a(n)$. For $X > 0$, we set

$$A^*(X) = \sum_{n \le X}{}^{*} a(n),$$

where the $*$ indicates that the last term is multiplied by $\frac{1}{2}$ if X is an integer. This slight modification of $A(X)$ accounts for the fact that Mellin inversion returns the average of the left-hand and right-hand limits at a jump discontinuity. We would like to relate $A^*(X)$ to the inverse Mellin transform of $D(s)$. We will prove several variants of this basic idea. The first being (classical) **Perron's formula** which is a consequence of Abel summation and Mellin inversion applied to Dirichlet series.

Theorem (Perron's formula, classical). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Then for any $c > \sigma_a$, we have*

$$A^*(X) = \frac{1}{2\pi i}\int_{(c)} D(s)X^s \,\frac{ds}{s}.$$

Proof. Let $\sigma > \sigma_a$. By Proposition 1.6, $A(X) \ll_\varepsilon X^{\sigma_a+\varepsilon}$. Taking ε sufficiently small, we find that $A(X)X^{-s} \to 0$ as $X \to \infty$. Abel summation then gives

$$D(s) = s\int_1^\infty A(u)u^{-(s+1)}\,du. \tag{1}$$

As $A(u) = 0$ in the interval $[0, 1)$, we can write the previous identity in the form

$$\frac{D(s)}{s} = \int_0^\infty A(u)u^{-(s+1)}\,du.$$

Mellin inversion immediately gives the result. □

In the proof of classical Perron's formula, we obtained a useful integral representation for a Dirichlet series. We collect this in the following corollary:

Corollary 1.11. *Let $D(s)$ be a Dirichlet series with finite and nonnegative abscissa of absolute convergence σ_a. Then for $\sigma > \sigma_a$, we have*

$$D(s) = s\int_1^\infty A(u)u^{-(s+1)}\,du.$$

Proof. The identity is Equation (1). $\square$

Classical Perron's formula is not always useful in applications because often it is necessary to estimate the integral of the Dirichlet series. As the integral need not be absolutely convergent, this would require nontrivial estimates for the Dirichlet series in vertical strips. Fortunately, there are two ways to correct for this defect each of which leads to a different variant of Perron's formula. The first is to truncate the integral while the latter is to introduce a smoothing function. In the former case, a careful estimation of the error term is often necessary while the latter case requires estimates for the smoothing function. Both variants can be equally useful and the choice of which to use is often dependent upon what methods are available in the current setting.

Let us first focus on the truncated variant. To state it, we will need to setup some notation and prove a lemma. For $c > 0$ and $T > 0$, consider the integrals

$$\delta(y) = \frac{1}{2\pi i}\int_{(c)} y^s\,\frac{ds}{s} \quad \text{and} \quad I(y,T) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} y^s\,\frac{ds}{s},$$

defined for $y > 0$. Note that $I(y,T)$ is just $d(y)$ truncated outside height T. The lemma we require gives an explicit evaluation of $\delta(y)$ and gives an approximation for the error between $\delta(y)$ and its truncation $I(y,T)$. The proof is quite laborious but standard.

Lemma 1.12. *We have*

$$\delta(y) = \begin{cases} 0 & \text{if } y < 1, \\ \frac{1}{2} & \text{if } y = 1, \\ 1 & \text{if } y > 1. \end{cases}$$

Moreover,

$$I(y,T) - \delta(y) = \begin{cases} O\left(y^c \min\left(1, \frac{1}{T\log(y)}\right)\right) & \text{if } y \neq 1, \\ O\left(\frac{1}{T}\right) & \text{if } y = 1. \end{cases}$$

Proof. Since $I(y,T) \to \delta(y)$ as $T \to \infty$, it suffices to estimate $I(y,T)$ and then take the limit as $T \to \infty$. First consider the case $y = 1$. A short computation shows

$$I(1,T) = \frac{1}{\pi}\tan^{-1}\left(\frac{T}{c}\right).$$

Truncate the Laurent series of the inverse tangent after the first term to write $\tan^{-1}(t) = \frac{\pi}{2} + O\left(\frac{1}{t}\right)$. Then

$$I(1,T) = \frac{1}{2} + O\left(\frac{1}{T}\right),$$

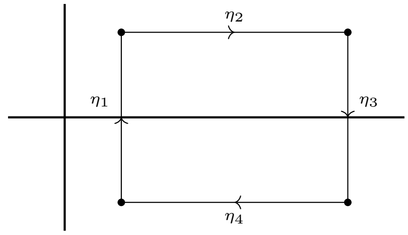

Figure 1: A rectangular contour.

and we see that $\delta(y) = \frac{1}{2}$. This proves everything when $y = 1$. Now suppose $y < 1$ and let $d > c$. Let $\eta = \eta_1 + \cdots + \eta_4$ be the rectangular contour in Figure 1 where the horizontal lines are along $t = \pm T$ and the vertical lines are along $\sigma = c$ and $\sigma = d$.
Consider

$$\frac{1}{2\pi i}\int_\eta y^s \frac{ds}{s}.$$

We will evaluate this integral in the limit as $d \to \infty$. The contour does not enclose the only pole of the integrand which is at $s = 0$. So on the one hand, the residue theorem gives

$$\frac{1}{2\pi i}\int_\eta y^s \frac{ds}{s} = 0.$$

On the other hand, the integral is a sum along all four contours. Observe that

$$\frac{1}{2\pi i}\int_{\eta_1} y^s \frac{ds}{s} = I(y,T).$$

For the integrals over η_2 and η_4, the parameterizations $s \mapsto \sigma \pm iT$ show

$$\frac{1}{2\pi i}\int_{\eta_2} y^s \frac{ds}{s} + \frac{1}{2\pi i}\int_{\eta_4} y^s \frac{ds}{s} = O\left(\frac{y^c}{\log(y)T}\right),$$

as $y < 1$. For the integral over η_3, the parameterization $s \mapsto d + it$ shows

$$\frac{1}{2\pi i}\int_{\eta_3} y^s \frac{ds}{s} = O\left(y^d \log(T)\right).$$

Whence

$$I(y,T) = O\left(\frac{y^c}{\log(y)T}\right),$$

upon taking the limit as $d \to \infty$ since $y < 1$. It follows that $\delta(y) = 0$. We now obtain another estimate for $I(y,T)$ this time using a modified contour. Let $\eta = \eta_1 + \eta_2$ be the semicircular contour in Figure 2 where the vertical line is along $\sigma = c$ with endpoints at $s = c \pm iT$.
As before, the residue theorem gives

$$\frac{1}{2\pi i}\int_\eta y^s \frac{ds}{s} = 0.$$

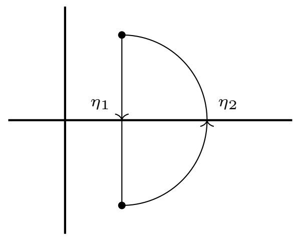

Figure 2: A semicircular contour.

However, now
$$\frac{1}{2\pi i}\int_{\eta_1} y^s \frac{ds}{s} = -I(y,T).$$
The parameterization $s \mapsto \sqrt{(c^2+T^2)}e^{i\theta}$ shows that
$$\frac{1}{2\pi i}\int_{\eta_2} y^s \frac{ds}{s} = O(y^c).$$
Whence
$$I(y,T) = O(y^c).$$
Combining these two estimates for $I(y,T)$ proves everything when $y < 1$. Now suppose $y > 1$ and let $d < 0$. We argue as in the case $y < 1$ except the rectangular contour is now as in Figure 3.

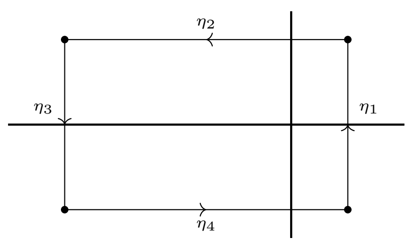

Figure 3: A rectangular contour.

This time the the contour encloses the simple pole of the integrand at $s = 0$ whose residue is 1. Arguing analogously, we find
$$I(y,T) = 1 + O\left(\frac{y^c}{\log(y)T}\right),$$
and so $\delta(y) = 1$. The corresponding semicircular contour is given in Figure 4.
This contour also encloses the simple pole of the integrand at $s = 0$ whose residue is 1. Arguing analogously, we obtain
$$I(y,T) = 1 + O(y^c).$$
Combining these two estimates for $I(y,T)$ proves everything when $y < 1$ and completes the proof. □

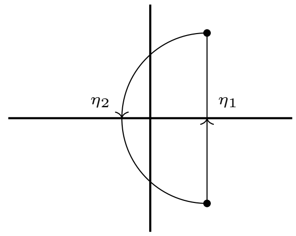

Figure 4: A semicircular contour.

This result will provide the estimate we need to prove (truncated) **Perron's formula**:

Theorem (Perron's formula, truncated). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Then for any $c > \sigma_a$ and $T > 0$, we have*

$$\begin{aligned}A^*(X) &= \frac{1}{2\pi i}\int_{c-iT}^{c+iT} D(s)X^s\,\frac{ds}{s}\\ &\quad + O\left(X^c \sum_{\substack{n\geq 1\\ n\neq X}} \frac{|a(n)|}{n^c}\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) + \delta_X|a(X)|\frac{1}{T}\right),\end{aligned}$$

where $\delta_X = 1, 0$ according to if X is an integer or not.

Proof. By Lemma 1.12, we may write

$$A^*(X) = \sum_{n\geq 1} a(n)\delta\left(\frac{X}{n}\right),$$

and

$$\delta(y) = I(y,T) + \begin{cases} O\left(y^c \min\left(1, \frac{1}{T\log(y)}\right)\right) & \text{if } y \neq 1,\\ O\left(\frac{1}{T}\right) & \text{if } y = 1.\end{cases}$$

Substituting the second identity into the first and combining the O-estimates gives

$$\begin{aligned}A^*(X) &= \sum_{n\geq 1} a(n)\frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{X^s}{n^s}\,\frac{ds}{s}\\ &\quad + O\left(X^c \sum_{\substack{n\geq 1\\ n\neq X}} \frac{|a(n)|}{n^c}\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) + \delta_X\frac{|a(X)|}{T}\right).\end{aligned}$$

As $D(s)$ converges absolutely, we may interchange the sum and the integral. The claim follows. ∎

Having discussed truncated Perron's formula, we turn to the variant where we instead introduce a smoothing function. We call $\psi(y)$ a **smooth weight** if it is a nonnegative bump function whose support is bounded away from zero. For any $X > 0$, we set

$$A_\psi(X) = \sum_{n \ge 1} a(n)\psi\left(\frac{n}{X}\right).$$

We distinguish between two cases. The first is when we choose $\psi(y)$ to assign weight 1 or 0 to the coefficients and we obtain sums such as

$$\sum_{\frac{X}{2} \le n < X} a(n) \quad \text{or} \quad \sum_{X \le n < X+Y} a(n).$$

Sums of this type are called **unweighted**. As an example of an unweighted sum, suppose $\psi(y)$ is a smooth weight that is identically 1 on $\left[\frac{1}{2}, 1\right]$ and supported in $\left[\frac{1}{2} - \frac{1}{X}, 1 + \frac{1}{X}\right]$. Then

$$A_\psi(X) = \sum_{\frac{X}{2} \le n \le X} a(n).$$

In the second case, we want $\psi(y)$ to dampen the coefficients with a weight other than 1 or 0. Sums of this type are called **weighted**. In either case, the Mellin transform $\Psi(s)$ of $\psi(y)$ is given by

$$\Psi(s) = \int_0^\infty \psi(y) y^s \, \frac{dy}{y}.$$

As $\psi(y)$ has compact support bounded away from zero, the integral defining $\Psi(s)$ is a locally absolutely uniformly convergent. In particular, $\Psi(s)$ is entire and Mellin inversion implies that $\psi(y)$ is the Mellin inverse of $\Psi(s)$. As for nice properties, $\Psi(s)$ exhibit rapid decay.

Proposition 1.13. *Suppose $\psi(y)$ is smooth weight and let $\Psi(s)$ be its Mellin transform. Then for bounded σ, we have*

$$\Psi(s) \ll (|s| + 1)^{-N},$$

for any positive integer N.

Proof. Consider

$$\Psi(s) = \int_0^\infty \psi(y) y^s \, \frac{dy}{y}.$$

Since $\psi(y)$ is compactly supported, integration by parts yields

$$\Psi(s) = \frac{1}{s} \int_0^\infty \psi'(y) y^{s+1} \, \frac{dy}{y}.$$

Repeated integration by parts gives

$$\Psi(s) = \frac{1}{s(s+1)\cdots(s+N-1)} \int_0^\infty \psi^{(N)}(y) y^{s+N} \, \frac{dy}{y}.$$

Therefore

$$\Psi(s) \ll (|s| + 1)^{-N} \int_0^\infty \psi^{(N)}(y) y^{\sigma+N} \, \frac{dy}{y}.$$

Now $\psi^{(N)}(y)$ is compactly supported in the same region as $\psi(y)$. In particular, $\psi^{(N)}(y)$ has compact support away from zero. Therefore

$$\int_0^\infty \psi^{(N)}(y)y^{\sigma+N}\,\frac{dy}{y} \ll 1,$$

and the estimate follows. $\square$

The following result is (smoothed) **Perron's formula**:

Theorem (Perron's formula, smoothed). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Let $\psi(y)$ be a smooth weight and let $\Psi(s)$ be its Mellin transform. Then for any $c > \sigma_a$, we have*

$$A_\psi(X) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s)X^s\,ds.$$

In particular,

$$\sum_{n\geq 1} a(n)\psi(n) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s)\,ds.$$

Proof. By Mellin inversion, we may write

$$A_\psi(X) = \sum_{n\geq 1}\frac{a(n)}{2\pi i}\int_{(c)} \Psi(s)\left(\frac{n}{X}\right)^{-s}\,ds.$$

Using Proposition 1.13, the sum and integral may be interchanged by absolute convergence. This gives

$$A_\psi(X) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s)X^s\,ds,$$

which is the first statement. For the second, take $X = 1$. $\square$

The integral in smoothed Perron's formula is absolutely convergent (this permitted the interchange of the sum and integral). In practice, this means that the integral can be estimated directly and we do not need to truncate. The compensation for this is that we have introduced a weighting factor to the sum of coefficients.

2 L-functions

Throughout, $s = \sigma + it$ and $u = \tau + ir$ will stand for complex variables with σ, τ, t, and r real. We also write

$$(u)_k = u(u+1)\cdots(u+(k-1)),$$

for the Pochhammer symbol. We will make use of standard properties of the gamma function especially Stirling's formula as well as the Mellin transform and its inversion theorem. Also, all sums and products over zeros of a function will be counted with multiplicity and ordered symmetrically with respect to the size of the ordinate if they are not absolutely convergent.

2.1 Analytic Data

An (analytic) **L-function** $L(s)$ is a Dirichlet series

$$L(s) = \sum_{n\geq 1} \frac{a_L(n)}{n^s},$$

with coefficients $a_L(n) \in \mathbb{C}$ such that $a_L(1) = 1$ and satisfying the following properties:

Analyticity: There exists a nonnegative integer m_L such that the Dirichlet series of $L(s)$ is absolutely convergent for $\sigma > 1$ and admits meromorphic continuation to $\mathbb{C}$ with at most a pole at $s = 1$ of order m_L. Moreover, $(s-1)^{m_L} L(s)$ is order 1.

Functional Equation: There is a positive integer q_L, called the **conductor** of $L(s)$, a positive integer d_L called the **degree** of $L(s)$, complex numbers $(\mu_j)_{j\leq d_L}$ called the **gamma parameters** of $L(s)$ that are either real or occur in conjugate pairs and satisfy $\mathrm{Re}(\mu_j) \geq 0$, and a complex number ε_L called the **root number** of $L(s)$ with $|\varepsilon_L| = 1$, all of which determine functions

$$\gamma(s, L) = \pi^{-\frac{d_L s}{2}} \prod_j \Gamma\left(\frac{s+\mu_j}{2}\right) \quad \text{and} \quad \Lambda(s, L) = q_L^{\frac{s}{2}} \gamma(s, L) L(s),$$

called the **gamma factor** of $L(s)$ and the **completion** of $L(s)$ respectively, where the latter admits meromorphic continuation to $\mathbb{C}$ with at most poles at $s = 0$ and $s = 1$ and satisfies

$$\Lambda(s, L) = \varepsilon_L \overline{\Lambda(1-\overline{s}, L)}.$$

This is called the **functional equation** for $L(s)$. The tuple $(\varepsilon_L, q_L, \mu_1, \ldots, \mu_{d_L})$ is called the **functional equation data** of $L(s)$.

Euler product: For every prime p, there exist complex numbers $(\alpha_j(p))_{j\leq d_L}$ called the **Euler parameters** of $L(s)$ at p satisfying $|\alpha_j(p)| \leq p^{\theta_L}$ for some $0 \leq \theta_L < 1$ and are such that $L(s)$ admits the Euler product

$$L(s) = \prod_p (1 - \alpha_1(p)p^{-s})^{-1} \cdots (1 - \alpha_{d_L}(p)p^{-s})^{-1},$$

for $\sigma > 1$. The polynomial L_p defined by

$$L_p(T) = (1 - \alpha_1(p)T) \cdots (1 - \alpha_{d_L}(p)T),$$

is called the **Euler factor** of $L(s)$ at p. If $p \nmid q_L$ then L_p is of degree d_L while if $p \mid q_L$ then L_p is of degree less than d_L. In the former case we say p is **unramified** while in the latter we say p is **ramified**.

An L-function is said to be **Selberg class** if it satisfies the following additional property:

Ramanujan bound: We may take $\theta_L = 0$ so that $|\alpha_j(p)| \leq 1$.

Some comments on the definition of L-functions are in order.

(i) As the order of the pole m_L is nonnegative, it is assumed that $L(1) \neq 0$ if $m_L = 0$.

(ii) As the Dirichlet series of $L(s)$ converges absolutely for $\sigma > 1$ its converges locally absolutely uniformly in this half-plane and therefore defines a holomorphic function there. Moreover, the Euler product necessarily converges locally absolutely uniformly in the same half-plane and hence defines a holomorphic function there as well which agrees with the Dirichlet series.

(iii) The bound $\mathrm{Re}(\mu_j) \geq 0$ ensures that the gamma factor is holomorphic in the half-plane $\sigma > 0$. As this factor is then guaranteed to be finite and nonzero at $s = 1$, the order of the pole at $s = 1$ of the completion is m_L. By the functional equation, order of the pole at $s = 0$ is also m_L.

(iv) If we merely assume $(s-1)^{m_L} L(s)$ is finite order then the functional equation forces the order to be 1. This can be deduced by considering the entire function

$$s^{m_L}(s-1)^{m_L}\Lambda(s, L),$$

and applying the Phragmén-Lindelöf convexity principle in vertical strips.

(v) The condition that the root number satisfies $|\varepsilon_L| = 1$ isn't strictly necessary. For applying the functional equation twice shows $|\varepsilon_L|^2 = 1$.

(vi) As the gamma function is conjugation-equivariant and the gamma parameters are real or occur in conjugate pairs, the gamma factor is also conjugation-equivariant. This means that we can write the functional equation in the form

$$q_L^{\frac{s}{2}}\gamma(s, L)L(s) = \varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s, L)\overline{L(1-\overline{s})}.$$

(vii) The Euler product implies that the coefficients $a_L(n)$ are multiplicative and are determined on prime powers by

$$a_L(p^k) = \sum_{j_1 \leq \cdots \leq j_k} \alpha_{j_1}(p) \cdots \alpha_{j_k}(p).$$

In other words, $a_L(p^k)$ is the complete symmetric polynomial of degree k in the Euler parameters at p. If the Ramanujan bound holds, then it follows that $a_L(n) \ll \sigma_{d_L}(n)$. This implies the slightly weaker, but more practical, estimate $a_L(n) \ll_\varepsilon n^\varepsilon$.

We will also define the **dual** $\overline{L}(s)$ of $L(s)$ by

$$\overline{L}(s) = \overline{L(\overline{s})}.$$

L-functions are closed under duals. Indeed, $\overline{L}(s)$ is an L-function with coefficients $a_{\overline{L}}(n) = \overline{a_L(n)}$, a pole at $s = 1$ of order $m_{\overline{L}} = m_L$, conductor $q_{\overline{L}} = q_L$, degree $d_{\overline{L}} = d_L$, gamma parameters $(\overline{\mu_j})_{j \leq d_L}$, root number $\varepsilon_{\overline{L}} = \overline{\varepsilon_L}$, gamma factor $\gamma(s, \overline{L}) = \gamma(s, L)$, completion

$\Lambda(s,\overline{L}) = \overline{\Lambda(\overline{s},L)}$, and Euler parameters $(\overline{\alpha_j(p)})_{j\le d_L}$ at p. Moreover, the dual of $L(s)$ is Selberg class if $L(s)$ is. We also say $L(s)$ is **self-dual** if it is equal to its dual. This is equivalent to the coefficients $a_L(n)$ being real. In any case, we can express the functional equation as

$$\Lambda(s,L) = \varepsilon_L \Lambda(1-s,\overline{L}),$$

so the functional equation relates an L-function with its dual. This formula will also be useful when taking derivatives of the functional equation.

It is clear from the definition that L-functions are closed under multiplication and that the degree is additive. Moreover, this closure respects Selberg class L-functions. This makes the set of L-functions into a graded commutative semigroup where the grading is induced by degree. It follows that there exist irreducible elements with respect to the grading and we say that an L-function is **primitive** if it such an irreducible. Clearly any L-function of degree 1 is primitive. Moreover, every L-function factors into a product of primitive L-functions. Such a factorization is only conjectured to be unique for Selberg class L-functions. A sufficient condition would be **Selberg's orthogonality conjecture**.

Conjecture (Selberg's orthogonality conjecture). *For any two primitive Selberg class L-functions $L_1(s)$ and $L_2(s)$, we have*

$$\sum_{p\le x} \frac{a_{L_1}(p)\overline{a_{L_2}(p)}}{p} = \delta_{L_1,L_2} \log\log(x) + O(1).$$

Indeed, we have the following result:

Proposition 2.1. *Assume Selberg's orthogonality conjecture. Then every Selberg class L-function factors uniquely into a product of primitive Selberg class L-functions.*

Proof. If the factorization into primitive Selberg-class L-functions were not unique, then we would have distinct factorizations satisfying

$$L_1(s)\cdots L_n(s) = M_1(s)\cdots M_m(s),$$

for some primitive Selberg class L-functions L_i and M_j. We may assume all of these are distinct for otherwise, we may cancel the common L-functions on either side. By uniqueness of coefficients of Dirichlet series, namely Proposition 1.3, compare the p-th coefficient to see that

$$\sum_i a_{L_i}(p) = \sum_j a_{M_j}(p).$$

Now consider

$$\sum_{p\le x} \frac{1}{p}\left(\sum_i a_{L_i}(p)\right)\left(\sum_j \overline{a_{M_j}(p)}\right) = \sum_{i,j}\sum_{p\le x} \frac{a_{L_i}(p)\overline{a_{M_j}(p)}}{p}.$$

By Selberg's orthogonality conjecture, the left-hand side is $\log\log(x)+O(1)$ while the right-hand side is $O(1)$. This is impossible and thus the factorization is unique. □

To an L-function $L(s)$, we associate its **analytic conductor** $\mathfrak{q}(s,L)$ defined by

$$\mathfrak{q}(s,L) = q_L \mathfrak{q}_\infty(s,L),$$

where

$$\mathfrak{q}_\infty(s,L) = \prod_j (|s+\mu_j|+3).$$

For convenience, we will also set

$$\mathfrak{q}(L) = \mathfrak{q}(0,L) \quad \text{and} \quad \mathfrak{q}_\infty(L) = \mathfrak{q}_\infty(0,L).$$

The choice of 3 in $|s+\mu_j|+3$ is a matter of convenience as could use any positive constant. In particular, it is useful when taking logarithms as $\log(|s+\mu_j|+3) \geq 1$. Estimates for the analytic conductor follow from those of the gamma function. Recall from Stirling's formula that

$$\Gamma(s) \ll_\varepsilon (|t|+3)^{\sigma-\frac{1}{2}} e^{-\frac{\pi}{2}|t|} \quad \text{and} \quad \frac{1}{\Gamma(s)} \ll (|t|+3)^{\frac{1}{2}-\sigma} e^{\frac{\pi}{2}|t|}, \tag{2}$$

for bounded σ provided that in the former estimate s is at least distance ε away from the poles of the gamma function. Hence

$$\frac{\Gamma(1-s)}{\Gamma(s)} \ll_\varepsilon (|t|+3)^{1-2\sigma},$$

for bounded σ provided s is at least distance ε away from the poles of $\Gamma(1-s)$. Then the estimates

$$\frac{\gamma(1-s,L)}{\gamma(s,L)} \ll_\varepsilon \mathfrak{q}_\infty(s,L)^{\frac{1}{2}-\sigma} \quad \text{and} \quad q_L^{\frac{1}{2}-s}\frac{\gamma(1-s,L)}{\gamma(s,L)} \ll_\varepsilon \mathfrak{q}(s,L)^{\frac{1}{2}-\sigma}, \tag{3}$$

hold for bounded σ provided s is at least distance ε away from the poles of $\gamma(1-s,L)$. An associated estimate can be obtain for the logarithmic derivative of the analytic conductor. Recall from the logarithm of Stirling's formula that

$$\frac{\Gamma'}{\Gamma}(s) \ll \log(|s|+3),$$

provided σ is bounded and s is at least distance ε away from the poles of the gamma function. Then the estimates

$$\frac{\gamma'}{\gamma}(s,L) \ll_\varepsilon \log \mathfrak{q}_\infty(s,L) \quad \text{and} \quad \log(q_L) + \frac{\gamma'}{\gamma}(s,L) \ll_\varepsilon \log \mathfrak{q}(s,L), \tag{4}$$

hold for bounded σ provided s is at least distance ε away from the poles of the gamma factor.

The **critical strip** of an L-function is the vertical strip left invariant by the transformation $s \mapsto 1-s$. This region can also be described as

$$\left\{ s \in \mathbb{C} : \left|\sigma - \frac{1}{2}\right| \leq \frac{1}{2} \right\}.$$

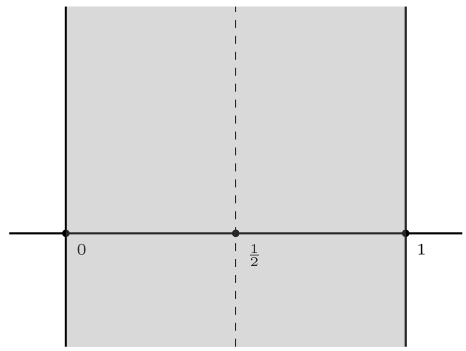

Figure 5: The critical strip.

The **critical line** is the vertical line left invariant by the transformation $s \mapsto 1-s$ which is also given by $\sigma = \frac{1}{2}$. The critical line bisects the critical strip vertically. The **central point** is the fixed point of the transformation $s \mapsto 1-s$, in other words, the point $s = \frac{1}{2}$. Clearly the central point is also the center of the critical line. The critical strip, critical line, and central point are all displayed in Figure 5.

In the half-plane $\sigma > 1$ we may study the analytic properties of $L(s)$ via its Dirichlet series. Using the functional equation, we may write

$$L(s) = \varepsilon_L q_L^{\frac{1}{2}-s} \frac{\gamma(1-s,L)}{\gamma(s,L)} \overline{L}(1-s).$$

This permits the study of $L(s)$ in the half-plane $\sigma < 0$ by using the Dirichlet series of $\overline{L}(1-s)$ and the the functional equation data. The interior of the critical strip is exactly the region where we cannot use either of these methods to study the analytic properties of $L(s)$. Of course, anything may be possible on the boundary lines $\sigma = 0$ and $\sigma = 1$ (for example Landau's theorem).

2.2 The Approximate Functional Equation

Despite not being able to study an L-function $L(s)$ in the critical strip by means of Dirichlet series, there is formula which acts as a compromise between the functional equation and the Dirichlet series. This formula is known as the approximate functional equation. The usefulness comes from the fact that the approximate functional equation is valid inside of the critical strip and therefore can be used to obtain analytic properties about $L(s)$ in that region. We will first derive a preliminary result showing $L(s)$ has at most polynomial growth.

Proposition 2.2. *Let $L(s)$ be an L-function with σ bounded and s at least distance ε away from the possible pole at $s = 1$. Then there is a positive constant A such that*

$$L(s) \ll_\varepsilon (|t|+3)^A.$$

Proof. Observe that $(s-1)^{m_L} L(s) \ll_\varepsilon (|t|+3)^{m_L}$ on the vertical line $\sigma = 1+\varepsilon$. On the vertical line $\sigma = -\varepsilon$, the functional equation and Equation (3) together show that there is a positive constant A'' for which

$$(s-1)^{m_L} L(s) \ll_\varepsilon (|t|+3)^{A''}.$$

As $(s-1)^{m_L}L(s)$ is entire and of order 1, we can apply the Phragmén-Lindelöf convexity principle to $(s-1)^{m_L}L(s)$ the vertical strip $-\varepsilon \le \sigma \le 1+\varepsilon$. Whence there is a positive constant A' such that

$$(s-1)^{m_L}L(s) \ll_\varepsilon (|t|+3)^{A'},$$

provided σ is bounded. Assuming s is at least distance ε away from the possible pole at $s=1$, diving by $(s-1)^{m_L}$ completes the proof. □

We are almost ready to prove the approximate function equation for an L-function $L(s)$. The formula itself consists of two sums representing the Dirichlet series at s and a dualized Dirichlet series at $1-s$ as well as a potential residue term. The dual sum comes equip with a term containing the data of the functional equation. Both sums will also be dampened by a smooth cutoff function. In the statement of the approximate function equation, we will make use of a test function $\Phi(u)$. We require $\Phi(u)$ be an even holomorphic function bounded in the vertical strip $|\tau| < c+1$ for some $c > 1$ and such that $\Phi(0) = 1$. For s in the critical strip, we let $V_s(y, L)$ be the inverse Mellin transform

$$V_s(y, L) = \frac{1}{2\pi i}\int_{(c)} \frac{\gamma(s+u, L)}{\gamma(s, L)}\Phi(u)y^{-u}\frac{du}{u},$$

defined for $y > 0$. Stirling's formula implies

$$\frac{\Gamma(s+u)}{\Gamma(s)} \ll_\varepsilon \frac{(|t+r|+3)^{\sigma+\tau-\frac{1}{2}}}{(|t|+3)^{\sigma-\frac{1}{2}}}e^{-\frac{\pi}{2}(|t+r|-|t|)},$$

for s in the critical strip and bounded τ provided $s+u$ is at least distance ε away from the poles of the gamma function. Whence

$$\frac{\gamma(s+u, L)}{\gamma(s, L)} \ll_\varepsilon \frac{\mathfrak{q}_\infty(s+u, L)^{\frac{\sigma+\tau}{2}-\frac{1}{4}}}{\mathfrak{q}_\infty(s, L)^{\frac{\sigma}{2}-\frac{1}{4}}}e^{-d_L\frac{\pi}{2}(|t+r|-|t|)}, \tag{5}$$

for s in the critical strip and bounded τ provided $s+u$ is at least distance ε away from the poles of the gamma factor. Since $\Phi(u)$ is bounded in the vertical strip $|\tau| < c+1$, the integrand exhibits exponential decay. Therefore the integral is locally absolutely uniformly convergent and hence $V_s(y, L)$ is smooth. The function $V_s(y, L)$ is the smooth cutoff function mentioned previously. We will also let $\varepsilon(s, L)$ be given by

$$\varepsilon(s, L) = \varepsilon_L q_L^{\frac{1}{2}-s}\frac{\gamma(1-s, L)}{\gamma(s, L)}.$$

This term appears as a factor in the dual sum accounting for the functional equation data. Lastly, we set

$$R(s, X, L) = \operatorname*{Res}_{u=1-s}\frac{\Lambda(s+u, L)\Phi(u)X^u}{u} + \operatorname*{Res}_{u=-s}\frac{\Lambda(s+u, L)\Phi(u)X^u}{u}.$$

This is the potential residue term that appears in the approximate functional equation. Observe that it is zero provided $s \neq 0, 1$. We now prove the **approximate function equation** for $L(s)$.

Theorem (Approximate functional equation). *Let $L(s)$ be an L-function, let $\Phi(u)$ be an even holomorphic function bounded in the vertical strip $|\tau| < c+1$ for some $c > 1$ and such that $\Phi(0) = 1$, and let $X \geq 1$. Then for s in the critical strip, we have*

$$L(s) = \sum_{n\geq 1} \frac{a_L(n)}{n^s} V_s\left(\frac{n}{\sqrt{q_L}X}, L\right) + \varepsilon(s, L) \sum_{n\geq 1} \frac{\overline{a_L(n)}}{n^{1-s}} V_{1-s}\left(\frac{nX}{\sqrt{q_L}}, L\right) + \frac{R(s, X, L)}{q_L^{\frac{s}{2}}\gamma(s, L)}.$$

Proof. Consider the integral

$$\frac{1}{2\pi i}\int_{(c)} \Lambda(s+u, L)\Phi(u)X^u \, \frac{du}{u}.$$

Stirling's formula shows that $\gamma(s+u, L)$ exhibits exponential decay while $L(s+u)$ has at most polynomial growth by Proposition 2.2. Since $\Phi(u)$ is bounded in the vertical strip $|\tau| < c+1$, it follows that the integrand exhibits exponential decay. Therefore the integral is locally absolutely uniformly convergent. We will evaluate the integral in two ways. On the one hand, we can expand $L(s+u)$ inside the integrand as a Dirichlet series and by absolute boundedness of the integral we may interchange the sum and integral. A short computation shows that this is

$$q_L^{\frac{s}{2}}\gamma(s, L) \sum_{n\geq 1} \frac{a_L(n)}{n^s} V_s\left(\frac{n}{\sqrt{q_L}X}, L\right).$$

On the other hand, we can shift the line of integration to $(-c)$. In doing so we pass by a simple pole at $u = 0$ and possible poles at $u = 1-s$ and $u = -s$ giving

$$\frac{1}{2\pi i}\int_{(-c)} \Lambda(s+u, L)\Phi(u)X^u \, \frac{du}{u} + \Lambda(s, L) + R(s, X, L).$$

Apply the functional equation and perform the change of variables $u \mapsto -u$ to rewrite this as

$$-\varepsilon_L \frac{1}{2\pi i}\int_{(c)} \Lambda(1-(s+u), \overline{L})\Phi(u)X^{-u} \, \frac{du}{u} + \Lambda(s, L) + R(s, X, L).$$

Analogous to the above, we can now expand $\overline{L}(1-(s+u))$ inside the integrand as a Dirichlet series and by absolute boundedness of the integral we may interchange the sum and integral. A short computation shows that our previous expression becomes

$$-\varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s, L) \sum_{n\geq 1} \frac{\overline{a_L(n)}}{n^{1-s}} V_s\left(\frac{nX}{\sqrt{q_L}}, L\right) + \Lambda(s, L) + R(s, X, L).$$

Equating these two evaluations and isolating the completion gives

$$\begin{aligned}\Lambda(s, L) = q_L^{\frac{s}{2}}\gamma(s, L) &\sum_{n\geq 1} \frac{a_L(n)}{n^s} V_s\left(\frac{n}{\sqrt{q_L}X}, L\right) \\ &+ \varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s, L) \sum_{n\geq 1} \frac{\overline{a_L(n)}}{n^{1-s}} V_{1-s}\left(\frac{nX}{\sqrt{q_L}}, L\right) + R(s, X, L).\end{aligned}$$

Diving by $q_L^{\frac{s}{2}}\gamma(s, L)$ completes the proof. $\square$

Let us now show how $V_s(y,L)$ has the effect of dampening the two dual sums appearing on the right-hand side of the approximate functional equation. In practice, it is common to choose $\Phi(u)$ such that it has exponential decay and we can make the vertical strip on which it is bounded arbitrarily wide. For example, let

$$\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right),$$

for some positive integer M. Clearly $\Phi(u)$ is an even holomorphic function in the vertical strip $|\tau| < (2M-1)+1$ and satisfies $\Phi(0) = 1$. In view of the identity $\cos(u) = \frac{e^{iu}+e^{-iu}}{2}$, we find that

$$\cos^{-4d_L M}\left(\frac{\pi u}{4M}\right) \ll_\varepsilon e^{-d_L \pi |r|}, \tag{6}$$

for $|\tau| < (2M-1)+1$ provided u is at least distance ε away from the poles of $\Phi(u)$. Therefore $\Phi(u)$ admits exponential decay. For this choice of $\Phi(u)$, $V_s(y,L)$ and its derivatives will possess rapid decay.

Proposition 2.3. *Let $L(s)$ be an L-function, set $\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right)$ for some positive integer M, in the inverse Mellin transform $V_s(y,L)$. Then for s in the critical strip, any nonnegative integer k, and positive integer N with $N < 2M$, $V_s(y,L)$ satisfies the estimates*

$$(-y)^k V_s^{(k)}(y,L) = \begin{cases} \delta_{k,0} + O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^N\right) & \text{if } y \ll \sqrt{\mathfrak{q}_\infty(s,L)}, \\ O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}\right) & \text{if } y \gg \sqrt{\mathfrak{q}_\infty(s,L)}. \end{cases}$$

In particular,

$$(-y)^k V_s^{(k)}(y,L) \ll_\varepsilon \left(1 + \frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}.$$

Proof. As we have already seen, the integrand defining $V_s(y,L)$ admits exponential decay. This permits us to differentiate under the integral sign and shift the line of integration. The former shows

$$(-y)^k V_s^{(k)}(y,L) = \frac{1}{2\pi i}\int_{(c)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u)(u)_k y^{-u}\frac{du}{u},$$

for any $1 < c < 2M$. Shifting to $(-N)$, we pass by a simple pole at $u = 0$ of residue 1 if and only if $k = 0$. This gives

$$(-y)^k V_s^{(k)}(y,L) = \delta_{k,0} + \frac{1}{2\pi i}\int_{(-N)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u)(u)_k y^{-u}\frac{du}{u}.$$

This integrand also exhibits exponential decay since the Pochhammer symbol grows polynomially. Therefore the integral is dominated by the contribution when $u \ll 1$ and the Equations (5) and (6) together show

$$(-y)^k V_s^{(k)}(y,L) = \delta_{k,0} + O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^N\right).$$

If we instead shift to (N), we do not pass by any poles and obtain

$$(-y)^k V_s^{(k)}(y,L) = \frac{1}{2\pi i}\int_{(N)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u)(u)_k y^{-u}\frac{du}{u}.$$

An analogous argument to estimate the remaining integral shows

$$(-y)^k V_s^{(k)}(y,L) = O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}\right).$$

From these O-estimates we obtain nontrivial bounds in the ranges $y \ll \sqrt{\mathfrak{q}_\infty(s,L)}$ and $y \gg \sqrt{\mathfrak{q}_\infty(s,L)}$ respectively. Combining both of these estimates produces the bound

$$(-y)^k V_s^{(k)}(y,L) \ll_\varepsilon \left(1+\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}.$$

□

With our choice of $\Phi(u)$, this result shows that $V_s(y,L)$ is essentially 1 for $y \ll \sqrt{\mathfrak{q}_\infty(s,L)}$ and has rapid decay thereafter as we can take M (and hence N) to be arbitrarily large.

2.3 The Riemann Hypothesis and Nontrivial Zeros

The zeros of an L-functions $L(s)$ has interesting behavior which are classified into two categories: trivial and nontrivial zeros. To describe them, recall that the completion

$$\Lambda(s,L) = q_L^{\frac{s}{2}}\gamma(s,L)L(s),$$

admits meromorphic continuation to $\mathbb{C}$ with at most poles at $s=0$ and $s=1$. As the gamma factor is never zero, any zero of the completion is necessarily a zero of $L(s)$. A **trivial zero** of $L(s)$ is any zero that is not a zero of the completion. Accordingly, this means that the trivial zeros are also poles of $\gamma(s,L)$ and as such must live in the region $\sigma \le 0$. They are all of the form $s = -(\mu_j + 2n)$ for some gamma parameter μ_j and nonnegative integer n. A **nontrivial zero** of $L(s)$ is any zero that is also a zero of the completion. Then the zeros of the completion are exactly the nontrivial zeros of $L(s)$. As $\gamma(s,L)L(s)$ is holomorphic and nonzero for $\sigma > 1$, the completion is nonzero in this region. By the functional equation, the completion is also nonzero for $\sigma < 0$. This means that all of the zeros of the completion occur in the critical strip which is to say that the nontrivial zeros belong to the critical strip. Let ρ be a nontrivial zero of $L(s)$. Then the functional equation implies that $1-\overline{\rho}$ is also a nontrivial zero of $L(s)$. This means that nontrivial zeros occur in pairs

$$\rho \quad \text{and} \quad 1-\overline{\rho}.$$

It is possible to say more when $L(s)$ takes real values for $s > 1$. For in this case, the Schwarz reflection principle implies $L(\overline{s}) = \overline{L(s)}$ and that $L(s)$ takes real values on the entire real line save for the possible pole at $s=1$. It follows from the functional equation that $\overline{\rho}$ and $1-\rho$ are also nontrivial zeros. Therefore the nontrivial zeros of $L(s)$ come in sets of four

$$\rho, \quad \overline{\rho}, \quad 1-\rho, \quad \text{and} \quad 1-\overline{\rho},$$

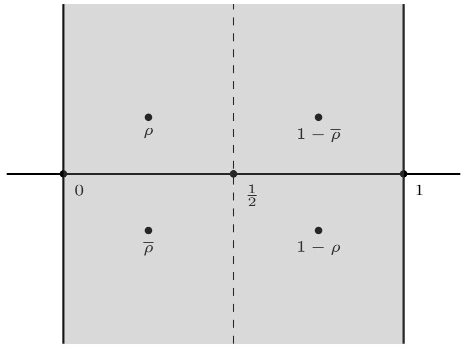

Figure 6: Symmetric nontrivial zeros.

as displayed in Figure 6.

The **Riemann hypothesis** for $L(s)$ says that this symmetry should be as simple as possible.

Conjecture (Riemann hypothesis). *The nontrivial zeros of the L-function $L(s)$ lie on the vertical line $\sigma = \frac{1}{2}$.*

While stated as a conjecture, we not expect the Riemann hypothesis to hold for just any L-function. We do however expect it to hold for Selberg class L-functions. In particular, the **Selberg class Riemann hypothesis** says that this symmetry should hold for any Selberg class L-function:

Conjecture (Selberg class Riemann hypothesis). *The nontrivial zeros of any Selberg class L-function $L(s)$ lie on the vertical line $\sigma = \frac{1}{2}$.*

So far, the Riemann hypothesis remains completely out of reach for any L-function and thus the Selberg class Riemann hypothesis does as well.

2.4 The Lindelöf Hypothesis and Moments

Instead of asking about the zeros of an L-function $L(s)$ on the critical line we can ask about its growth there. Let us begin this investigation by deriving an upper bound for $L(s)$ on the critical line using the Phragmén-Lindelöf convexity principle. This amounts to a more refined version of the argument used in Proposition 2.2 for the critical strip.

Theorem 2.4. *Let $L(s)$ be an L-function. Then for $-\varepsilon \leq \sigma \leq 1+\varepsilon$, we have*

$$L(s) \ll_\varepsilon \mathfrak{q}(s,L)^{\frac{1-\sigma}{2}+\varepsilon},$$

provided s is at least distance ε away from the possible pole at $s=1$.

Proof. Since $(s-1)^{m_L}L(s)$ is entire and of order 1, we will apply the Phragmén-Lindelöf convexity principle just outside the edges of the critical strip. Just past the right edge along the vertical line $\sigma = 1+\varepsilon$, we have

$$(s-1)^{m_L}L(s) \ll_\varepsilon (s-1)^{m_L}.$$

Just past the left edge along the vertical line $\sigma = -\varepsilon$, the functional equation and Equation (3) together imply the bound

$$(s-1)^{m_L} L(s) \ll_\varepsilon (s-1)^{m_L} \mathfrak{q}(s, L)^{\frac{1}{2}+\varepsilon}.$$

By the Phragmén-Lindelöf convexity principle, we obtain

$$(s-1)^{m_L} L(s) \ll_\varepsilon (s-1)^{m_L} \mathfrak{q}(s, L)^{\frac{1-\sigma}{2}+\varepsilon},$$

provided $-\varepsilon \leq \sigma \leq 1+\varepsilon$. Assuming s is at least distance ε away from the possible pole at $s = 1$, dividing by $(s-1)^{m_L}$ completes the proof. $\square$

At the critical line this theorem produces the bound

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon}.$$

This is known as the **convexity bound** for $L(s)$. The **Lindelöf hypothesis** for $L(s)$ says that the exponent can be reduced to ε:

Conjecture (Lindelöf hypothesis). *The L-function $L(s)$ satisfies*

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\varepsilon}.$$

Unlike the Riemann hypothesis, the Lindelöf hypothesis is expected to hold for any L-function. In any case, the **Selberg class Lindelöf hypothesis** says that the exponent can be reduced to ε for any Selberg class L-function:

Conjecture (Selberg class Lindelöf hypothesis). *Any Selberg class L-function $L(s)$ satisfies*

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\varepsilon}.$$

Like the Riemann hypothesis, we have been unable to prove the Lindelöf hypothesis for any L-function. However, in practice the Lindelöf hypothesis is more tractable. Generally speaking, any improvement upon the exponent in the convexity bound in any aspect of the analytic conductor is called a **subconvexity estimate** (or a **convexity breaking bound**). In the uniform case, we would like to prove bounds of the form

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\delta+\varepsilon},$$

for some $0 \leq \delta \leq \frac{1}{4}$. The convexity bound says that we may take $\delta = \frac{1}{4}$ while the Lindelöf hypothesis for $L(s)$ implies that we may take $\delta = 0$. Any uniform subconvexity estimate would give a δ strictly less than $\frac{1}{4}$. Subconvexity estimates are viewed with great interest. This is primarily because of their connection to the Lindelöf hypothesis, but also because any improvement upon the convexity bound in any aspect of the analytic conductor often has drastic consequences in applications.

Remark 2.5. Some subconvexity bounds are deserving of names. The cases $\delta = \frac{3}{16}$ and $\delta = \frac{1}{6}$ are referred to as the **Burgess bound** and **Weyl bound** respectively.

With a little more work, we can obtain a a similar bound for the derivatives of L-functions. First observe that Theorem 2.4 gives the estimate

$$L(s) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon},$$

in the vertical strip $\left|\sigma - \frac{1}{2}\right| \leq \frac{\varepsilon}{2}$. This is just slightly stronger than the convexity bound. By Cauchy's integral formula, we may write

$$L^{(k)}\left(\frac{1}{2}+it\right) = \frac{k!}{2\pi i}\int_\eta \frac{L(s)}{(s-\frac{1}{2}-it)^{k+1}}\,ds,$$

where η is the circle about $\frac{1}{2}+it$ of radius $\frac{\varepsilon}{2}$. The parameterization $u \mapsto \frac{1}{2}+it+\frac{\varepsilon}{2}e^{i\theta}$ for η shows that

$$L^{(k)}\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon}.$$

This is known as the **convexity bound** for $L^{(k)}(s)$.

While both the Riemann and Lindelöf hypotheses remain out of reach for any L-function $L(s)$, the latter is more tractable because the convexity bound gives a weak estimate. Unfortunately, breaking the convexity bound for any L-function is a very daunting task requiring extremely modern techniques. This approach of breaking convexity to study the Lindelöf hypothesis is quite direct. However, there is an indirect approach by considering averages of the L-function in question. This is the study of moments of L-functions. For any L-function $L(s)$ and positive integer k, we define the k-th **moment** of $L(s)$, namely $M_k(T,L)$, by

$$M_k(T,L) = \int_0^T \left|L\left(\frac{1}{2}+it\right)\right|^k dt,$$

for any positive T. It is useful to think of $M_k(T,L)$ as a k-th power average of $L\left(\frac{1}{2}+it\right)$. One of the fundamental reasons why the study of moments is useful is that sufficient estimates for $M_{2k}(T,L)$ in T is equivalent to the Lindelöf hypothesis.

Proposition 2.6. *Let $L(s)$ be an L-function. The truth of the Lindelöf hypothesis for $L(s)$ is equivalent to the estimate*

$$M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon},$$

for all positive integers k.

Proof. First suppose the Lindelöf hypothesis for $L(s)$ holds. Then we obtain the estimate

$$M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon},$$

at once for all k. Now suppose that $M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon}$ for all k. If the Lindelöf hypothesis for $L(s)$ is false then there is some $\delta > 0$ and positive unbounded sequence $(t_n)_{n\geq 1}$ such that

$$\left|L\left(\frac{1}{2}+it_n\right)\right| > C\mathfrak{q}\left(\frac{1}{2}+it_n, L\right)^{\delta},$$

for any positive constant C. In particular, we may write

$$\left|L\left(\frac{1}{2}+it_n\right)\right| > Ct_n^{d_L\delta}.$$

The convexity bound for $L(s)$ implies the weaker estimate $\left|L'\left(\frac{1}{2}+it\right)\right| \le C't^{d_L}$ for some positive constant C' provided t is bounded away from zero. Now write

$$\left|L\left(\frac{1}{2}+it\right)-L\left(\frac{1}{2}+it_n\right)\right| = \left|\int_{t_n}^{t} L'\left(\frac{1}{2}+ir\right)\,dr\right|.$$

Taking $C = 2C'$, a short computation using these results shows that

$$\left|L\left(\frac{1}{2}+it\right)-L\left(\frac{1}{2}+it_n\right)\right| \le \frac{C}{2}t^{d_L\delta},$$

provided t is bounded away from zero and satisfies $|t-t_n| \le t_n^{d_L(\delta-1)}$. For such t, it follows that

$$\left|L\left(\frac{1}{2}+it\right)\right| > \frac{C}{2}t_n^{d_L\delta}.$$

Now take $T = \frac{4}{3}t_n$ so that the interval $(t_n - t_n^{d_L(\delta-1)}, t_n + t_n^{d_L(\delta-1)})$ is contained in $\left(\frac{T}{2}, T\right)$. Then

$$M_{2k}(T,L) > 2\left(\frac{C}{2}\right)^{2k}\left(\frac{3}{4}\right)^{(2k+1)d_L\delta-d_L} T^{(2k+1)d_L\delta-d_L}.$$

This is a contradiction for sufficiently large k which completes the proof. □

The usefulness of this result is that the difficulty of proving the Lindelöf hypothesis for $L(s)$ has been transferred to proving sufficient asymptotics for the moments $M_{2k}(T,L)$ each of which should, heuristically speaking, be an easier problem to resolve on its own.

2.5 Estimates on the Critical Line

We now turn to the question of how to obtain estimates on the critical line. There are many methods one can apply often depending upon the particular L-function of interest. Here we will provide a method that works in vast generality and reduces the estimation of $L\left(\frac{1}{2}+it\right)$ to that of estimating smoothed finite sums of coefficients. To achieve this we need to establish the existence of certain bump functions. We say that a function $\omega(y)$ is a **smooth dyadic weight** if it is a nonnegative bump function supported in $[1-\delta, 2+\delta]$ and such that

$$\sum_{k\in\mathbb{Z}} \omega\left(\frac{y}{2^k}\right) = 1,$$

for $y > 0$. This last condition means $\left(\omega\left(\frac{y}{2^k}\right)\right)_{k\in\mathbb{Z}}$ is a partition of unity on $(0,\infty)$. To see that dyadic weights exist, let $\psi(y)$ be a smooth weight that is identically 1 on $[1,2]$ and supported in $[1-\delta, 2+\delta]$. Write

$$\sigma(y) = \sum_{k\in\mathbb{Z}} \psi\left(\frac{y}{2^k}\right).$$

Then $\sigma(y)$ is finite as finitely many of its summands are nonzero since any positive y satisfies $2^k \leq y < 2^{k+1}$ for some unique integer k. This forces $\sigma(y)$ to be smooth and bounded. It is also bounded away from zero as $\sigma(y) \geq 1$ because $\psi(y)$ is identically 1 on $[1,2]$. Then we may take

$$\omega(y) = \frac{\psi(y)}{\sigma(y)}.$$

This shows that smooth dyadic weights exist. For any $t \in \mathbb{R}$ and $X > 0$, set

$$A_\omega(t, X, L) = \sum_{n\geq 1} a_L(n) n^{-it} \omega\left(\frac{n}{X}\right).$$

For convenience, we will also let

$$A_\omega(X, L) = A_\omega(0, X, L).$$

Our result estimates $L(s)$ at $s = \frac{1}{2} + it$ in terms of $A_\omega(t, X, L)$.

Theorem 2.7. *Let $L(s)$ be an L-function and let $\omega(y)$ be a smooth dyadic weight. Then*

$$L\left(\frac{1}{2} + it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2} + it, L\right)^\varepsilon \sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it, L\right)}} \left(\frac{|A_\omega(t, X, L)|}{\sqrt{X}}\right).$$

Proof. Take $s = \frac{1}{2} + it$, $X = 1$, and $\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right)$ for some large positive integer M in the approximate functional equation. This permits us to write

$$\begin{aligned} L\left(\frac{1}{2} + it\right) &= \sum_{n\geq 1} \frac{a_L(n) n^{-it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right) \\ &\quad + \varepsilon\left(\frac{1}{2} + it, L\right) \sum_{n\geq 1} \frac{\overline{a_L(n)} n^{it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right) + \frac{R\left(\frac{1}{2} + it, 1, L\right)}{q_L^{\frac{1}{4}+i\frac{t}{2}} \gamma\left(\frac{1}{2} + it, L\right)}. \end{aligned}$$

Equation (3) implies $\varepsilon\left(\frac{1}{2} + it, L\right) \ll 1$ while Equations (2) and (6) together imply that the residue term exhibits exponential decay. This implies that right-hand side of the approximate functional equation is

$$O\left(\sum_{n\geq 1} \frac{a_L(n) n^{-it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right)\right) + O\left(\sum_{n\geq 1} \frac{\overline{a_L(n)} n^{it}}{\sqrt{n}} V_{\frac{1}{2}-it}\left(\frac{n}{\sqrt{q_L}}, L\right)\right) + O(e^{-|t|}).$$

We now turn to estimating the remaining sums which will be accomplished by applying the smooth dyadic weight. Consider the first sum. By our choice of $\Phi(u)$, the estimates in Proposition 2.3 are valid which implies that the summands exhibit rapid decay for $n \gg \sqrt{\mathfrak{q}\left(\frac{1}{2} + it, L\right)}$. Whence the sum is say

$$\sum_{n \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it, L\right)}} \frac{a_L(n) n^{-it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right) + O_\varepsilon\left(\mathfrak{q}\left(\frac{1}{2} + it, L\right)^\varepsilon\right).$$

Now write

$$V_s(y,L) = \sum_{k\in\mathbb{Z}} \omega\left(\frac{y}{2^k}\right) V_s(y,L).$$

Apply this identity to the sum and interchange the resulting two sums by local finiteness to obtain

$$\sum_{k\in\mathbb{Z}} \sum_{n\ll\sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}} \frac{a_L(n)n^{-it}}{\sqrt{n}}\omega\left(\frac{n}{2^k\sqrt{q_L}}\right) V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right).$$

By compact support of the smooth dyadic weight, for each k the inner sum over n is supported on the dyadic block $n \asymp X_k$ where $X_k = 2^k\sqrt{q_L}$. As $n \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}$, the summands which contribute must satisfy $k \ll \log \mathfrak{q}_\infty\left(\frac{1}{2}+it,L\right)$. Therefore our double sum can be expressed as

$$\sum_{k\ll\log\mathfrak{q}_\infty\left(\frac{1}{2}+it,L\right)} \sum_{n\asymp X_k} \frac{a_L(n)n^{-it}}{\sqrt{n}}\omega\left(\frac{n}{X_k}\right) V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right).$$

Let $V(y) = \frac{1}{\sqrt{y}}V_{\frac{1}{2}+it}\left(\frac{y}{\sqrt{q_L}},L\right)$. By Abel summation, the inner sum is O of

$$\sup_{n\asymp X_k} |A_\omega(X_k,t,L)|\,|V(n)| + \int_{u\asymp X_k} |A_\omega(u,t,L)||V'(u)|\,du.$$

In view of Proposition 2.3, we have the estimates

$$V(y) = O\left(\frac{1}{\sqrt{y}}\right) \quad \text{and} \quad V'(y) = O\left(\frac{1}{y^{\frac{3}{2}}}\right),$$

for $y \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}$. Whence our previous expression is O of

$$\sup_{X\asymp X_k}\left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right).$$

As there are $O_\varepsilon\left(\mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon\right)$ many such k, the bound above implies that our sum is O_ε of

$$\mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon \sup_{X\ll\sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}}\left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right).$$

The estimate for the second sum is treated analogously. Thus

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon \sup_{X\ll\sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}}\left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right),$$

upon absorbing smaller order O-estimates. $\square$

If $A_\omega(t,X,L) \ll_\varepsilon X^{\frac{1}{2}+\delta+\varepsilon}$ for some $\delta \geq 0$ then the supremum in Theorem 2.8 will automatically pick out the largest possible bound when X is the square root of the analytic conductor. This means that the savings of $\sqrt{X}$ in the denominator is essentially as large as possible. For example, the convexity bound follows upon showing that we may take $\delta = \frac{1}{2}$. Since the Dirichlet series of $L(s)$ is absolutely convergent for $\sigma < 1$, we have a bound of shape

$$\sum_{n \leq X} |a_L(n)| \ll_\varepsilon X^{1+\varepsilon}.$$

Then $A_\omega(t,X,L) \ll_\varepsilon X^{1+\varepsilon}$ as well and we obtain

$$\sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}} \frac{|A_\omega(t,X,L)|}{\sqrt{X}} \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it,L\right)^{\frac{1}{4}+\varepsilon}, \tag{7}$$

which gives the convexity bound. As we believe $L(s)$ satisfies the Lindelöf hypothesis, we should expect lots of cancellation in the sum $A_\omega(t,X,L)$. In fact, the expected cancellation should be $O\left(\mathfrak{q}\left(\frac{1}{2}+it,L\right)^{\frac{1}{4}}\right)$ in light of the aforementioned bound. This means we expect $A_\omega(t,X,L)$ to exhibit square-root cancellation in the sense

$$A_\omega(t,X,L) \ll_\varepsilon X^{\frac{1}{2}+\varepsilon},$$

which is to say we may take $\delta = 0$. As the coefficients $a_L(n)$ may very well be real, the expected cancellation must come from the terms $n^{-it} = e^{-it\log(n)}$ which act on the coefficients $a_L(n)$ by rotation. In effect, we should think of the terms $a_L(n)n^{-it}$ as modeling identically distributed random variables. In order to exploit this cancellation via analytic methods, one can make use of smoothed Perron's formula. Letting $\Omega(s)$ be the Mellin transform of $\omega(y)$, smoothed Perron's formula gives

$$A_\omega(t,X,L) = \frac{1}{2\pi i}\int_{(c)} L\left(u+it\right)\Omega(u)X^u\,du,$$

for any $c > 1$. Then estimates for $A_\omega(t,X,L)$ can be obtained from those of this inverse Mellin transform.

The case when $t = 0$ deserves special interest as we are estimating the value of an L-function at the central point. The value of $L(s)$ at the central point is called the **central value** of $L(s)$. Many important properties about an L-function can be connected to its central value. Any argument used to estimate the central value of an L-function is called a **central value estimate**. Taking $t = 0$ in Theorem 2.7 gives a way to obtain central value estimates.

Theorem 2.8. *Let $L(s)$ be an L-function and let $\omega(y)$ be a smooth dyadic weight. Then*

$$L\left(\frac{1}{2}\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2},L\right)^\varepsilon \max_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2},L\right)}} \left(\frac{|A_\omega(X,L)|}{\sqrt{X}}\right).$$

Proof. Take $t = 0$ in Theorem 2.7. □

2.6 Logarithmic Derivatives

Recall that $L(s)$ nonzero in the half-plane $\sigma > 1$. As this region is simply connected, there exists a unique holomorphic logarithm $\log L(s)$ there. By absolute convergence of the Euler product, we may write

$$\log L(s) = -\sum_p \sum_j \log(1 - \alpha_j(p)p^{-s}).$$

Moreover, the bound $|\alpha_j(p)| \le p^{\theta_L}$ ensures that $|\alpha_j(p)p^{-s}| < 1$ in this half-plane. Therefore the Taylor series of the logarithm is valid in this region and we may further write

$$\log L(s) = \sum_p \sum_j \sum_{k\ge1} \frac{\alpha_j(p)^k}{kp^{ks}}.$$

While this triple sum converges for $\sigma > 1$, the bound $|\alpha_j(p)p^{-s}| < p^{\theta_L - \sigma}$ only guarantees absolutely convergence for $\sigma > 1 + \theta_L$. It is in this half-plane that we may differentiate termwise. A short computation shows

$$\frac{L'}{L}(s) = -\sum_{n\ge1} \frac{\Lambda_L(n)}{n^s},$$

where

$$\Lambda_L(n) = \begin{cases} \sum_j \alpha_j(p)^k \log(p) & \text{if } n = p^k, \\ 0 & \text{otherwise.} \end{cases}$$

We call $\Lambda_L(n)$ the **von Mangoldt coefficients** of $L(s)$. Also observe that the dual satisfies $\Lambda_{\overline{L}}(n) = \overline{\Lambda_L(n)}$.

There is an incredibly useful formula for the logarithmic derivative of $L(s)$ which is often the starting point for deeper analytic investigations. To deduce it, we will work with a slightly modified version of the completion of $L(s)$. Define $\xi(s, L)$ by

$$\xi(s, L) = (s(1-s))^{m_L} \Lambda(s, L).$$

Then $\xi(s, L)$ is just the completion with the potential poles at $s = 0$ and $s = 1$ removed so that it is nonzero there. In particular, $\xi(s, L)$ is entire and all of its zeros are exactly the nontrivial zeros of $L(s)$. Also observe that the dual satisfies $\xi(s, \overline{L}) = \overline{\xi(\overline{s}, L)}$ and the functional equation implies

$$\xi(s, L) = \varepsilon_L \xi(1 - s, \overline{L}),$$

which we refer to as the **functional equation** for $\xi(s, L)$. Also, $\xi(s, L)$ is of order 1 by Stirling's formula and that $(s-1)^{m_L} L(s)$ is of order 1. Then the Hadamard factorization theorem implies

$$\xi(s, L) = e^{A_L + B_L s} \prod_\rho \left(1 - \frac{s}{\rho}\right) e^{\frac{s}{\rho}},$$

for some constants A_L and B_L and that the sum

$$\sum_\rho \frac{1}{|\rho|^{1+\varepsilon}}, \tag{8}$$

is convergent. Our next result will utilize the Hadamard factorization of $\xi(s, L)$ to produce a formula relating the logarithmic derivative of $L(s)$ to its zeros.

Proposition 2.9. *For any L-function $L(s)$, we have*

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s, L) - B_L - \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

Proof. We will compute the logarithmic derivative of $\xi(s, L)$ in two ways. On the one hand, taking the logarithmic derivative of the definition of $\xi(s, L)$ yields

$$\frac{\xi'}{\xi}(s, L) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s, L) + \frac{L'}{L}(s).$$

On the other hand, taking the logarithmic derivative of the Hadamard factorization gives

$$\frac{\xi'}{\xi}(s, L) = B_L + \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

Equating these expressions to obtain

$$B_L + \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s, L) + \frac{L'}{L}(s).$$

This is equivalent to the desired formula. □

Explicit evaluation of the constants A_L and B_L can be challenging and heavily depend upon the particular L-function under investigation. However, useful formulas are not too difficult to obtain. For example, B_L is related to a sum over nontrivial zeros. To see this, using the Hadamard factorization of $\xi(s, L)$ to take the logarithmic derivative of both sides of the functional equation gives

$$B_L + \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) = -\overline{B_L} + \sum_{\rho}\left(\frac{1}{s-(1-\overline{\rho})} - \frac{1}{\overline{\rho}}\right).$$

As the nontrivial zeros occur in pairs ρ and $1-\overline{\rho}$, isolating B_L shows that

$$\mathrm{Re}(B_L) = -\sum_{\rho}\mathrm{Re}\left(\frac{1}{\rho}\right). \tag{9}$$

2.7 Zero Density Estimates

The distribution of the zeros of an L-function is central to how the L-function encodes arithmetic information about its coefficients. Here we introduce a method of counting zeros of L-functions which gives a simple understanding of their density. We first require an immensely useful lemma.

Lemma 2.10. *Let $L(s)$ be an L-function. Then the following properties hold:*

(i) *For any real T, we have*

$$\sum_{\rho} \frac{1}{1+(T-\gamma)^2} = O(\log \mathfrak{q}(T,L)).$$

In particular, $O(\log \mathfrak{q}(T,L))$ many nontrivial zeros of $L(s)$ satisfy $|T-\gamma| \leq 1$.

(ii) *Let $s = \sigma + iT$ for any real T. Then*

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\substack{\rho \\ |T-\gamma|\leq 1}} \frac{1}{s-\rho} + O_\varepsilon(\log \mathfrak{q}(T,L)),$$

provided σ is bounded, and s is at least distance ε away from the nontrivial zeros of $L(s)$ and the poles of the gamma factor, and $s \neq 0, 1$.

Proof. We will prove the properties separately.

(i) Let $s = \sigma + iT$ for some $\sigma > 1 + \theta_L$ so that the logarithmic derivative of $L(s)$ is absolutely bounded. Taking the real part of the formula in Proposition 2.9 and using Equations (4) and (9) together gives

$$\sum_{\rho} \mathrm{Re}\left(\frac{1}{s-\rho}\right) = O(\log \mathfrak{q}(T,L)),$$

upon absorbing all other terms into the O-estimate. For our choice of s, we have

$$\mathrm{Re}\left(\frac{1}{s-\rho}\right) \gg \frac{1}{1+(T-\gamma)^2},$$

which implies the first statement. The second statement follows from the first as all of the terms in the sum are positive and at least $\frac{1}{2}$.

(ii) Let $s = \sigma + iT$ with σ bounded. Also suppose s is at least distance ε away from the nontrivial zeros of $L(s)$ and the poles of the gamma factor and $s \neq 0, 1$. Invoking Proposition 2.9 and using Equation (4) gives

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) + O_\varepsilon(\log \mathfrak{q}(T,L)).$$

A short computation shows

$$\frac{1}{s-\rho} + \frac{1}{\rho} \ll \frac{1}{|\gamma|^2} + \frac{1}{|T-\gamma|^2},$$

and it follows that

$$\sum_{\substack{\rho \\ |T-\gamma|>1}} \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) \ll \sum_{\substack{\rho \\ |T-\gamma|>1}} \frac{1}{|\gamma|^2} + \sum_{\substack{\rho \\ |T-\gamma|>1}} \frac{1}{|T-\gamma|^2}.$$

The first sum on the right-hand side is $O(1)$ by Equation (8). As we are restricting to nontrivial zeros such that $|T-\gamma|>1$ in the second sum, it is $O(\log \mathfrak{q}(T,L))$ by (i). Also, (i) implies

$$\sum_{\substack{\rho \\ |T-\gamma|\le 1}} \frac{1}{\rho} = O_\varepsilon(\log \mathfrak{q}(T,L)).$$

Altogether, this shows

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\substack{\rho \\ |T-\gamma|\le 1}} \frac{1}{s-\rho} + O_\varepsilon(\log \mathfrak{q}(T,L)),$$

as claimed. □

With this lemma we can deduce a result which estimates the number of nontrivial zeros of $L(s)$ in a box. For any $T\ge 0$, define

$$N(T,L) = |\{\rho=\beta+i\gamma \in \mathbb{C} : L(\rho)=0 \text{ with } 0\le\beta\le 1 \text{ and } |\gamma|\le T\}|.$$

In other words, $N(T,L)$ is the number of nontrivial zeros of $L(s)$ whose ordinate belongs to $[-T,T]$. We will use the argument principle to deduce an asymptotic formula for $N(T,L)$.

Theorem 2.11. *For any L-function $L(s)$ and $T\ge 1$, we have*

$$N(T,L) = \frac{T}{\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right) + O(\log \mathfrak{q}(T,L)).$$

In particular,

$$\frac{N(T,L)}{T} \sim \frac{1}{\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right).$$

Proof. Let $T\ge 1$ and set

$$N'(T,L) = \left|\{\rho=\beta+i\gamma \in \mathbb{C} : L(\rho)=0 \text{ with } 0\le\beta\le 1 \text{ and } 0<\gamma\le T\}\right|.$$

As the nontrivial zeros occur in pairs ρ and $1-\overline{\rho}$, it follows that

$$N(T,L) = 2N'(T,L) + O(1),$$

where $O(1)$ accounts for the possible real nontrivial zeros of which there are finitely many. We will estimate $N'(T,L)$ and in doing so we may assume $L(s)$ does not have a nontrivial zero in the horizontal strip $T-\delta\le t\le T+\delta$ for some $\delta>0$ by varying T by a sufficiently small constant, if necessary, and observing that $N'(T,L)$ is modified by a quantity of size $O(\log \mathfrak{q}(T,L))$ by Lemma 2.10 (i). Taking δ smaller, if necessary, we may additionally assume $\Lambda(s,L)$ has no nontrivial zeros in the horizontal strip $-\delta\le t<0$. Let $\eta=\sum_i \eta_i$ be the rectangular contour in Figure 7 where the horizontal lines are along $t=T$ and $t=-\delta$ and the vertical lines are along $\sigma=1+\theta_L+\delta$ and $\sigma=-(\theta_L+\delta)$.

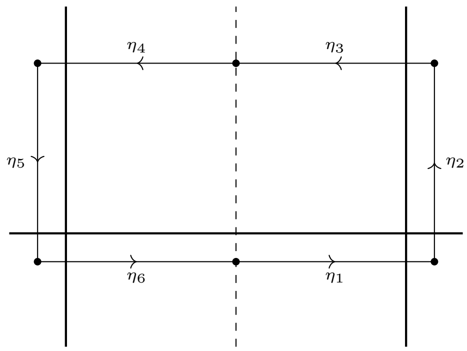

Figure 7: A zero counting contour.

By our previous comments and the argument principle, we may write

$$N'(T,L) = \frac{1}{2\pi i}\int_\eta \frac{\xi'}{\xi}(s,L)\,ds + O(\log \mathfrak{q}(T,L)),$$

where

$$\frac{1}{2\pi i}\int_\eta \frac{\xi'}{\xi}(s,L)\,ds = \frac{1}{2\pi}\Delta_\eta \arg \xi(s,L).$$

For convenience, set $\eta_R = \eta_1 + \eta_2 + \eta_3$ and $\eta_L = \eta_4 + \eta_5 + \eta_6$. Observe that η_R is taken to $-\eta_L$ under the change of variables $s \mapsto 1 - \overline{s}$. Using this fact and the functional equation, a short computation shows

$$\Delta_{\eta_R} \arg \xi(s,L) = \Delta_{\eta_L} \arg \xi(s,L).$$

In other words, the change in argument along η_R is equal to the change in argument along η_L. Whence

$$\frac{1}{2\pi i}\int_\eta \frac{\xi'}{\xi}(s,L)\,ds = \frac{1}{\pi}\Delta_{\eta_R} \arg \xi(s,L).$$

We will now estimate the integral by estimating the change in argument along η_R of each factor in

$$\xi(s,L) = (s(1-s))^{m_L} q_L^{\frac{s}{2}} \pi^{-\frac{d_L s}{2}} \prod_j \Gamma\left(\frac{s+\mu_j}{2}\right) L(s).$$

For the contribution from $(s(1-s))^{m_L}$, first recall that $\arg(s) = \tan^{-1}\left(\frac{t}{\sigma}\right)$ provided $\sigma > 0$. In this case, we have the estimate $\arg(s) = \frac{\pi}{2} + O\left(\frac{1}{t}\right)$ by truncating the Laurent series of the inverse tangent after the first term. A short computation shows

$$\Delta_{\eta_R} \arg((s(1-s))^{m_L}) = O\left(\frac{1}{T}\right).$$

For the remaining factors, we will use $\arg(s) = \operatorname{Im}(\log(s))$. The contributions from $q_L^{\frac{s}{2}}$ and $\pi^{-\frac{d_L s}{2}}$ are easily estimated to be

$$\Delta_{\eta_R} \arg\left(q_L^{\frac{s}{2}}\right) = \frac{T}{2}\log(q_L) + O(1),$$

and
$$\Delta_{\eta_R} \arg\left(\pi^{-\frac{d_L s}{2}}\right) = \frac{T}{2}\log\left(\frac{1}{\pi^{d_L}}\right) + O(1),$$
respectively. For the product of gamma functions, first observe that the logarithm of Stirling's formula gives
$$\log\Gamma(s) = \frac{1}{2}\log(2\pi) + \left(s - \frac{1}{2}\right)\log(s) - s + O\left(\frac{1}{s}\right),$$
provided $t \geq 1$ say. A short computation shows
$$\Delta_{\eta_R} \arg\Gamma\left(\frac{s+\mu_j}{2}\right) = \frac{T}{2}\log\left(\frac{T}{2e}\right) + O(1),$$
and it follows that the contribution from the product of gamma functions is
$$\Delta_{\eta_R} \arg\left(\prod_j \Gamma\left(\frac{s+\mu_j}{2}\right)\right) = \frac{T}{2}\log\left(\frac{T^{d_L}}{(2e)^{d_L}}\right) + O(1).$$
For $L(s)$, being by observing that
$$\Delta_{\eta_R} \arg(L(s)) = \operatorname{Im}\left(\int_{\eta_R} \frac{L'}{L}(s)\, ds\right).$$
As the logarithmic derivative of $L(s)$ is absolutely bounded for $\sigma = 1 + \theta_L + \delta$, the integral is $O(1)$ on η_2. On η_1 and η_3, use Lemma 2.10 (ii) to write
$$\operatorname{Im}\left(\int_{\eta_1\cup\eta_3} \frac{L'}{L}(s)\, ds\right) = \sum_{\substack{\rho \\ |T-\gamma|\leq 1}} \operatorname{Im}\left(\int_{\eta_1\cup\eta_3} \frac{1}{s-\rho}\, ds\right) + O(\log \mathfrak{q}(T,L)),$$
where we have absorbed the terms corresponding to 0 and 1 into the O-estimate. Each summand is $\Delta_{\eta_1\cup\eta_3}\arg(s-\rho) = O(1)$. As there are $O(\log\mathfrak{q}(T,L))$ many terms by Lemma 2.10 (i), it follows that
$$\Delta_{\eta_R} \arg(L(s)) = O(\log \mathfrak{q}(T,L)).$$
Combining our change in argument estimates gives
$$N'(T,L) = \frac{T}{2\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right) + O(\log\mathfrak{q}(T,L)),$$
from which the asymptotic formula follows. The asymptotic equivalence is immediate since it is a strictly weaker asymptotic. $\square$

It is worth noting that the main term in the asymptotic formula of the previous result comes from the change in argument of $q_L^{\frac{s}{2}}\gamma(s,L)$ along η_3 (and η_4 by the functional equation). In particular, the contribution from $L(s)$ is only to the error term. This is a good example of how information of an L-function is intrinsically connected to its gamma factor.

Also observe that the asymptotic equivalence can be interpreted as saying that for large T the density of $N(T,L)$ is approximately $\frac{1}{\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right)$. Since this grows as $T\to\infty$, we see that the nontrivial zeros tend to accumulate farther up the critical strip with logarithmic growth. To account for this accumulation, if $\rho=\beta+i\gamma$ is a nontrivial zero of $L(s)$ then we call $\rho_{\text{unf}}=\beta+i\omega$ the **unfolded nontrivial zero** corresponding to ρ where we set

$$\omega=\frac{\gamma}{\pi}\log\left(\frac{q_L|\gamma|^{d_L}}{(2\pi e)^{d_L}}\right).$$

For any $W\geq 0$, define

$$N_{\text{unf}}(W,L)=|\{\rho_{\text{unf}}=\beta+i\omega\in\mathbb{C}:L(\rho)=0\text{ with }0\leq\beta\leq 1\text{ and }|\omega|\leq W\}|.$$

In other words, $N_{\text{unf}}(T,L)$ is the number of unfolded nontrivial zeros of $L(s)$ whose ordinate belongs to $[-W,W]$. The point of considering unfolded nontrivial zeros is that they are uniformly dense along the critical strip.

Proposition 2.12. *For any L-function $L(s)$ and $W\geq\frac{1}{\pi}\log\left(\frac{q_L}{(2\pi e)^{d_L}}\right)$, we have*

$$\frac{N_{\mathit{unf}}(W,L)}{W}\sim 1.$$

Proof. Consider the function $f(t)$ defined by

$$f(t)=\frac{t}{\pi}\log\left(\frac{q_L|t|^{d_L}}{(2\pi e)^{d_L}}\right),$$

for real t. Since $f(t)$ is a strictly increasing continuous function, it has an inverse $g(w)$. It follows that $|\omega|\leq W$ if and only if $|\rho|\leq g(W)$ whence

$$N_{\text{unf}}(W,L)=N(g(W),L).$$

But by Theorem 2.11 and that $g(w)$ is the inverse of $f(t)$, we have $N(g(W),L)\sim W$. Then

$$N_{\text{unf}}(W,L)\sim W,$$

which is equivalent to the claim. $\square$

We interpret this result as saying that the unfolded nontrivial zeros are evenly spaced opposed to Theorem 2.11 which say that they tend to accumulate up the critical strip.

2.8 Zero-free Regions

Although the Riemann hypothesis for any L-function remains out of reach, some progress has been made to understand regions inside of the critical strip for which there are no nontrivial zeros except for possibly one real exception. Such a region is said to be a **zero-free region**. We will derive a standard zero-free region for any L-function under a mild assumption about the real part of its logarithmic derivative. First we require a lemma which bounds the possible amount of real zeros.

Lemma 2.13. *Let $L(s)$ be an L-function such that the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$. Then $L(1) \neq 0$ whence $m_L \geq 0$. Moreover, there exists a positive constant c such that $L(s)$ has at most m_L real zeros in the region*

$$\sigma \geq 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}.$$

Proof. Observe that the real parts of the terms corresponding to the nontrivial zeros in Lemma 2.10 (ii) are positive provided $\sigma > 1$. Letting $s = \sigma$ and taking the real part of this formula, we obtain the inequality

$$\sum_{\beta} \mathrm{Re}\left(\frac{1}{\sigma - \beta}\right) \leq \frac{m_L}{\sigma - 1} + \mathrm{Re}\left(\frac{L'}{L}(\sigma)\right) + O(\log \mathfrak{q}(L)),$$

upon discarding all of the terns corresponding to nontrivial zeros except those which are real and absorbing the term corresponding to 0 into the O-estimate. Also note that the left-hand side is positive. As the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$ by assumption, we further have

$$\sum_{\beta} \mathrm{Re}\left(\frac{1}{\sigma - \beta}\right) \leq \frac{m_L}{\sigma - 1} + O(\log \mathfrak{q}(L)).$$

From this inequality we see that $\beta \neq 1$. For if some $\beta = 1$, we have $m_L < 0$ whence the right-hand side is negative for σ sufficiently close to 1 contradicting positivity of the left-hand side. Thus $L(1) \neq 0$ and hence $m_L \geq 0$ proving the first statement.

For the second statement, recall that $L(s)$ is nonzero for $\sigma > 1$. Letting c be a positive constant, there are finitely many, say n, real zeros β satisfying

$$\beta \geq 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}.$$

Setting $\sigma = 1 + \frac{2c}{\log \mathfrak{q}(L)}$, a short computation using the previous two inequalities gives

$$\frac{n \log \mathfrak{q}(L)}{2c + \frac{c}{(m_L+1)}} \leq \left(\frac{m_L}{2c} + O(1)\right) \log \mathfrak{q}(L).$$

Isolating n yields

$$n \leq m_L + \frac{m_L}{2(m_L + 1)} + O(c).$$

Taking c smaller, if necessary, shows $n \leq m_L$. As $L(s)$ is nonzero for $\sigma > 1$, it follows that there are at most m_L real zeros in the region

$$\sigma \geq 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}. \qquad \square$$

We can now prove our zero-free region result.

Theorem 2.14. *Let $L(s)$ be an L-function such that the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$. Then there exists a positive constant c such that $L(s)$ has no zeros in the region*

$$\sigma \geq 1 - \frac{c}{\log(\mathfrak{q}(L)(|t|+3))},$$

except for at most m_L many real zeros β each with $\beta < 1$.

Proof. For real r, let $M(s)$ be the L-function defined by

$$M(s) = L^{m_L}(s)\overline{L}^{m_L}(s)L^{2m_L}(s+ir)\overline{L}^{2m_L}(s-ir).$$

Then $d_M = 6m_L d_L$ and $\mathfrak{q}(M)$ satisfies

$$\mathfrak{q}(M) < \mathfrak{q}(L)^{6m_L}(|r|+3)^{4m_L}.$$

As the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$, the same holds $M(s)$ as well. Now let $\rho = \beta + i\gamma$ be a complex nontrivial zero of $L(s)$. Setting $r = \gamma$, $M(s)$ has a real nontrivial zero at $s = \beta$ of multiplicity at least $4m_L$ and a pole at $s = 1$ of order at most $2m_L$. From Lemma 2.13, it follows that there is a positive constant c for which $M(s)$ can have at most $2m_L$ real nontrivial zeros in the region

$$\sigma \geq 1 - \frac{c}{\log(\mathfrak{q}(L)(|\gamma|+3))}.$$

Therefore the region

$$\sigma \geq 1 - \frac{c}{\log(\mathfrak{q}(L)(|t|+3))},$$

does not contain any complex nontrivial zeros of $L(s)$. Now let β be a real nontrivial zero of $L(s)$. Setting $r = 0$, $M(s)$ has a real nontrivial zero at $s = \beta$ of multiplicity at least $6m_L$ and a pole at $s = 1$ of order at most $6m_L^2$. Whence Lemma 2.13 implies upon taking c smaller, if necessary, that there are at most m_L many real zeros in this region. If such a real zero β exists then $\beta < 1$ because $L(s)$ is nonzero for $\sigma > 1$ and must have a pole at $s = 1$. This completes the proof. $\square$

Some comments about this result are in order. First observe that the larger c can be taken the closer the curve bounding the zero-free region approaches the critical line. As we do not have a nonzero lower bound for c, this constant could be quite small and so we are only guaranteed a zero-free region very close to the line $\sigma = 1$. This means that the possible real zeros could be very close to 1. Moreover, as the nontrivial zeros of $L(s)$ occur in pairs ρ and $1-\overline{\rho}$, the zero-free region implies a symmetric zero-free region about the critical line with at most one real zero in each half as displayed in Figure 8. It is possible to obtain stronger zero-free regions in certain cases but this is heavily dependent upon the particular L-function under investigation. In many cases it is possible to augment the proof of Theorem 2.14 to make the constant c effective and such results have important applications. A possible real zero β in Theorem 2.14 is referred to as a **Siegel zero**. Such real zeros are conjectured to not exist. For example, Siegel zeros are conjectured to not exist for Selberg class L-functions in view of the Selberg class Riemann hypothesis. Nevertheless, it is an interesting question to see how the presence of a Siegel zero can affect the behavior of the associated L-function.

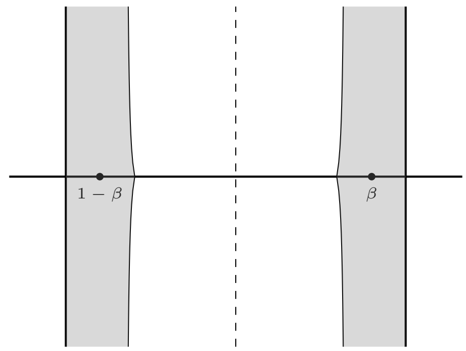

Figure 8: A symmetric zero-free region.

2.9 The Explicit Formula

There is a formula, somewhat analogous to the approximate functional equation, that relates a weighted sum of the von Mangoldt coefficients of any L-function $L(s)$ to a sum over its nontrivial zeros.

Theorem. *Let $\psi(y)$ be a smooth weight where $\Psi(s)$ is its Mellin transform and set $\phi(y) = y^{-1}\psi(y^{-1})$ so that its Mellin transform $\Phi(s)$ satisfies $\Phi(s) = \Psi(1-s)$. Then for any L-function $L(s)$, we have*

$$\sum_{n\geq 1}\Big(\Lambda_L(n)\psi(n) + \overline{\Lambda_L(n)}\phi(n)\Big) = \psi(1)\log(q_L) + m_L\Psi(1) - \sum_{\rho}\Psi(\rho)$$
$$+ \frac{1}{2\pi i}\int_{\left(\frac{1}{2}\right)}\left(\frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L)\right)\Psi(s)\,ds.$$

Proof. Smoothed Perron's formula implies

$$\sum_{n\geq 1}\Lambda_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(c)} -\frac{L'}{L}(s)\Phi(s)\,ds,$$

for any $c > 1 + \theta_L$. Let $\varepsilon > 0$ be such that there are no trivial zeros in the vertical strip $-2\varepsilon \leq \sigma < 0$. As the logarithmic derivative is absolutely bounded for $\sigma > 1 + \theta_L$, Proposition 2.2 implies that the integrand has rapid decay and therefore the integral is locally absolutely uniformly convergent. This permits us to shift the line of integration. Shifting to $(-\varepsilon)$, we pass by simple poles at $s = 1$ and $s = \rho$ for every nontrivial zero ρ obtaining

$$\sum_{n\geq 1}\Lambda_L(n)\psi(n) = m_L\Psi(1) - \sum_{\rho}\Psi(\rho) + \frac{1}{2\pi i}\int_{(-\varepsilon)} -\frac{L'}{L}(s)\Psi(s)\,ds.$$

Now take the logarithmic derivative of the functional equation to obtain

$$-\frac{L'}{L}(s) = \log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L) + \frac{\overline{L}'}{\overline{L}}(1-s).$$

Thus the remaining integral above may be expressed as

$$\frac{1}{2\pi i}\int_{(-\varepsilon)}\left(\log(q_L)+\frac{\gamma'}{\gamma}(s,L)+\frac{\gamma'}{\gamma}(1-s,L)+\frac{\overline{L}'}{\overline{L}}(1-s)\right)\Psi(s)\,ds.$$

The integral over the first term is $\psi(1)\log(q_L)$ by Mellin inversion. As for the last term, smoothed Perron's formula and that $\Phi(s)=\Psi(1-s)$ together give

$$\frac{1}{2\pi i}\int_{(-\varepsilon)}-\frac{\overline{L}'}{\overline{L}}(1-s)\Psi(s)\,ds=\sum_{n\geq 1}\overline{\Lambda_L(n)}\phi(n).$$

Whence

$$\sum_{n\geq 1}\left(\Lambda_L(n)\psi(n)+\overline{\Lambda_L(n)}\phi(n)\right)=\psi(1)\log(q_L)+m_L\Psi(1)-\sum_{\rho}\Psi(\rho)$$
$$+\frac{1}{2\pi i}\int_{(-\varepsilon)}\left(\frac{\gamma'}{\gamma}(s,L)+\frac{\gamma'}{\gamma}(1-s,L)\right)\Psi(s)\,ds.$$

The remaining integrand has rapid decay by Proposition 1.13 and Equation (4) and so the integral is locally absolutely uniformly convergent. Therefore we may shift the line of integration. Shifting to $\left(\frac{1}{2}\right)$, we pass over no poles. This completes the proof. $\square$

There is a much more useful variant which is obtained by applying truncated Perron's formula and performing various estimates. For this, we need to introduce two related functions. For any L-function $L(s)$, we define $\psi(x,L)$ by

$$\psi(X,L)=\sum_{n\leq X}\Lambda_L(n),$$

for $X>0$. In other words, $\psi(X,L)$ is just the sum of the von Mangoldt coefficients of $L(s)$ up to X. It turns out that many useful applications of L-functions can be derived from sufficiently nice estimates for $\psi(X,L)$. This is primarily due to the fact that $\psi(X,L)$ is closely related a sum over the nontrivial zeros of $L(s)$ which is strongly suggested by the previous result. Since we will be taking inverse Mellin transforms, it will be useful to work with a slightly modified version of $\psi(X,L)$. Accordingly, we define Define $\psi^*(x,L)$ by

$$\psi^*(X,L)=\sideset{}{^*}\sum_{n\leq X}\Lambda_L(n),$$

for $X>0$ and where the $*$ indicates that the last term is multiplied by $\frac{1}{2}$ if X is an integer. In fact, this difference is not apparent unless X is a prime power for otherwise $\Lambda_L(X)=0$. Nevertheless, $\psi^*(x,L)$ is exactly $\psi(x,L)$ except that its value is halfway between the left-hand and right-hand limits when X is a prime power. This modification is important because it will allow us to apply truncated Perron formulas directly. The relationship between $\psi^*(X,L)$ and the nontrivial zeros of $L(s)$ is known as the **explicit formula** for $L(s)$.

Theorem (Explict Formula). *For any L-function $L(s)$ and $T \geq X > 1$, we have*

$$\psi^*(X, L) = m_L X - \sum_{\substack{\rho \\ |\gamma| < T}} \frac{X^\rho}{\rho} + \sum_j \sum_{n \geq 0} \frac{X^{-(\mu_j + 2n)}}{\mu_j + 2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right) + O\left(\frac{X^c \log^2 \mathfrak{q}(T, L)}{T}\right).$$

In particular,

$$\psi^*(X, L) = m_L X - \sum_{\rho} \frac{X^\rho}{\rho} + \sum_j \sum_{n \geq 0} \frac{X^{-(\mu_j + 2n)}}{\mu_j + 2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right).$$

Proof. Truncated Perron's formula gives

$$\psi^*(X, L) - J(X, T) \ll X^c \sum_{\substack{n \geq 1 \\ n \neq X}} \frac{|\Lambda_L(n)|}{n^c} \min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) + \delta_X \frac{|\Lambda_L(X)|}{T},$$

where

$$J(X, T) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} -\frac{L'}{L}(s) X^s \frac{ds}{s},$$

for any $T > 0$, $c > 1 + \theta_L$, and $\delta_X = 0$ unless X is a prime power. We will now estimate the right-hand side above bounding the sum. In the ranges $n \ll X$ and $n \gg X$, we have $\log\left(\frac{X}{n}\right) \gg 1$ so that

$$\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \frac{1}{T}.$$

In the former case, the sum is finite so that the contribution of these n is

$$X^c \sum_{\substack{n \ll X \\ n \neq X}} \frac{|\Lambda_L(n)|}{n^c} \min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \frac{X^c}{T}.$$

In the latter case, we estimate the sum by absolute converge so that the contribution of these n is

$$X^c \sum_{\substack{n \gg X \\ n \neq X}} \frac{|\Lambda_L(n)|}{n^c} \min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \frac{X^c}{T}.$$

In the range $n \asymp X$, the Taylor expansion of the logarithm gives $\log\left(\frac{X}{n}\right) \ll \frac{X-n}{X}$ so that

$$\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \min\left(1, \frac{X}{T|X-n|}\right).$$

Whence

$$X^c \sum_{\substack{n \asymp X \\ n \neq X}} \frac{|\Lambda_L(n)|}{n^c} \min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \sum_{\substack{n \asymp X \\ |n-X| \le \frac{X}{T}}} |\Lambda_L(n)| + \frac{X}{T} \sum_{\substack{n \asymp X \\ |n-X| > \frac{X}{T}}} \frac{|\Lambda_L(n)|}{|X-n|}.$$

Using the estimate $\Lambda_L(n) \ll n^{\theta_L} \log(n)$ and approximating the last sum on the right-hand side by its corresponding integral, we obtain

$$X^c \sum_{\substack{n \asymp X \\ n \neq X}} \frac{|\Lambda_L(n)|}{n^c} \min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) \ll \frac{X^{1+\theta_L}\log(X)}{T} + \frac{X^{1+\theta_L}\log^2(X)}{T}.$$

The second term on the right-hand side dominates so in view of the previous estimates and the bound $\Lambda_L(X) \ll X^{\theta_L}\log(X)$ for the $\delta_X \frac{|\Lambda_L(X)|}{T}$ term, we have

$$\psi^*(X, L) - J(X, T) \ll \frac{X^c}{T} + \frac{X^{1+\theta_L}\log^2(X)}{T}.$$

We will now estimate the integral $J(X, T)$ by appealing to the residue theorem. Let T not coincide with the ordinate of a nontrivial zero. By Lemma 2.10 (i), the number of nontrivial zeros ρ of $L(s)$ satisfying $|\gamma \mp T| < 1$ is $O(\log \mathfrak{q}(T, L))$. Therefore, among the ordinates of these nontrivial zeros, there must be a gap of size $O\left(\frac{1}{\log \mathfrak{q}(T,L)}\right)$. So upon varying T by a bounded amount so that it belongs to this gap, we can ensure

$$\gamma \mp T \gg \frac{1}{\log \mathfrak{q}(T, L)},$$

for all nontrivial zeros ρ as they occur in pairs ρ and $1 - \overline{\rho}$. Also, choose a $U > c - 1$ with $U \gg T$ and such that $-U$ bounded away from the trivial zeros. Let $\eta = \sum_i \eta_i$ be the rectangular contour in Figure 9 where the horizontal lines are along $t = \pm T$ and the vertical lines are along $\sigma = c$ and $\sigma = -U$.

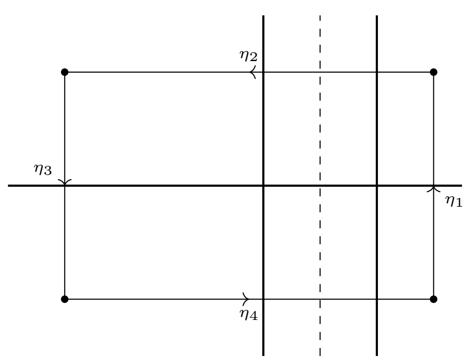

Figure 9: Contour for the explicit formula.

In view of Proposition 2.9, the contour encloses a simple pole of the integrand at $s = 1$ of residue $m_L X$, a simple pole at $s = \rho$ for every nontrivial zero ρ whose ordinate γ satisfies

$|\gamma| < T$ of residue $-\frac{X^\rho}{\rho}$, a simple pole at $s = -(\mu_j + 2n)$ for every gamma parameter μ_j and nonnegative integer n satisfying $n < \frac{U - \mathrm{Re}(\mu_j)}{2}$ of residue $\frac{X^{-(\mu_j+2n)}}{\mu_j+2n}$, and double or simple pole at $s = 0$ according to if $L(0) = 0$ or not. Thus

$$\frac{1}{2\pi i}\int_\eta -\frac{L'}{L}(s)X^s\,\frac{ds}{s} = m_L X - \sum_{\substack{\rho \\ |\gamma|<T}} \frac{X^\rho}{\rho} + \sum_j \sum_{0 \le n < \frac{U-\mathrm{Re}(\mu_j)}{2}} \frac{X^{-(\mu_j+2n)}}{\mu_j + 2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right).$$

On the other hand, the integral is a sum along its four contours. Observe that

$$J(X,T) = \frac{1}{2\pi i}\int_{\eta_1} -\frac{L'}{L}(s)X^s\,\frac{ds}{s}.$$

For the integrals along η_2 and η_4, we require distinct estimates when $1 - c \le \sigma \le c$ and $\sigma < 1 - c$. For the former, Lemma 2.10 (ii) gives

$$\frac{L'}{L}(s) = \sum_{|\gamma \mp T|<1} \frac{1}{s-\rho} + O(\log \mathfrak{q}(T,L)),$$

for $s = \sigma \pm iT$ with $1 - c \le \sigma \le c$. By our choice of T, $|s - \rho| \gg \frac{1}{\log \mathfrak{q}(T,L)}$ so that each term in the sum is $O(\log \mathfrak{q}(T,L))$. There are at most $O(\log \mathfrak{q}(T,L))$ such terms by Lemma 2.10 (i) and so

$$\frac{L'}{L}(s) = O(\log^2 \mathfrak{q}(T,L)),$$

in this region. For the latter, take the logarithmic derivative of the functional equation to obtain

$$-\frac{L'}{L}(s) = \log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L) + \frac{\overline{L}'}{\overline{L}}(1-s).$$

Let s be such that $\sigma < 1 - c$ and suppose s bounded away from the trivial zeros. Using Equation (4) and that the logarithmic derivative on the right-hand side converges absolutely, we obtain a bound

$$\frac{L'}{L}(s) = O(\log \mathfrak{q}(s,L)),$$

in this region. The previous two estimates imply that the integrals along η_2 and η_4 are O of

$$\frac{\log^2 \mathfrak{q}(T,L)}{T}\int_{1-c}^{c} X^\sigma\,d\sigma + \frac{\log \mathfrak{q}(T,L)}{T}\int_{-U}^{1-c} X^\sigma\,d\sigma.$$

Whence

$$\frac{1}{2\pi i}\int_{\eta_2 \cup \eta_4} -\frac{L'}{L}(s)X^s\,\frac{ds}{s} \ll \frac{X^c \log^2 \mathfrak{q}(T,L)}{T\log(X)} + \frac{X^{1-c}\log \mathfrak{q}(T,L)}{T\log(X)}.$$

As for the vertical integral along η_3, the previous estimate of the logarithmic derivative of $L(s)$ is also valid. Since $U \gg T$, it is O of

$$\frac{\log \mathfrak{q}(U,L)}{U}\int_{-T}^{T} x^{-U}\,dt.$$

Then

$$\frac{1}{2\pi i}\int_{\eta_3} -\frac{L'}{L}(s)X^s\,\frac{ds}{s} \ll \frac{T\log\mathfrak{q}(U,L)}{X^U U}.$$

Upon taking the limit as $U \to \infty$, this term tends to zero and we find that

$$J(X,T) = m_L X - \sum_{\substack{\rho \\ |\gamma|<T}} \frac{X^\rho}{\rho} + \sum_j \sum_{n\geq 0} \frac{X^{-(\mu_j+2n)}}{\mu_j+2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right) + O\left(\frac{X^c\log^2\mathfrak{q}(T,L)}{T\log(X)}\right),$$

because the sum over n converges absolutely which permits the interchange of limit and sum. Our estimate for $\psi^*(X,L)$ then becomes

$$\begin{aligned}\psi^*(X,L) = m_L X &- \sum_{\substack{\rho \\ |\gamma|<T}} \frac{X^\rho}{\rho} + \sum_j \sum_{n\geq 0} \frac{X^{-(\mu_j+2n)}}{\mu_j+2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right) \\ &+ O\left(\frac{X^c}{T} + \frac{X^{1+\theta_L}\log^2(X)}{T}\right) + O\left(\frac{X^c\log^2\mathfrak{q}(T,L)}{T\log(X)}\right).\end{aligned}$$

We now set $c = 1+\theta_L+\frac{1}{\log(X)}$ so that $X^c = eX^{1+\theta_L}$ and

$$\begin{aligned}\psi^*(X,L) = m_L X &- \sum_{\substack{\rho \\ |\gamma|<T}} \frac{X^\rho}{\rho} + \sum_j \sum_{n\geq 0} \frac{X^{-(\mu_j+2n)}}{\mu_j+2n} - \operatorname*{Res}_{s=0}\left(\frac{X^s}{s}\frac{L'}{L}(s)\right) \\ &+ O\left(\frac{X^c\log^2\mathfrak{q}(T,L)}{T}\right).\end{aligned}$$

This proves the first statement. Taking the limit as $T \to \infty$ proves the second. $\square$